85

Stem Cells, Kidney Regeneration, and Gene and Cell Therapy in Nephrology

Benjamin S. Freedman

CHAPTER OUTLINE

KEY POINTS

- Nephron loss is a major cause of chronic and end-stage kidney disease, which current therapies are insufficient to prevent.
- Recent advances in cell culture and reprogramming make it possible to derive new nephron progenitor cells from patients, which can be expanded and differentiated into kidney organoids.
- Stem cell therapies can be delivered by intravenous administration, cortical injection, or beneath the kidney capsule, but these methods require further optimization to demonstrate safety and efficacy.
- Larger functional units including metanephric implants, xenografts, and bioartificial kidneys are being tested for their capacity to substitute whole kidneys.
- Gene transfer techniques have improved in safety and efficacy, and are beginning to reach the clinic for nonkidney disorders, usually in combination with cell therapies.
- CRISPR and related technologies enable precision editing of genetic elements and control of gene expression, but remain relatively inefficient and unpredictable in therapeutic settings.
- Combinatorial use of stem cell therapy and gene therapy will provide an expanded menu of treatment options for the nephrologist of the future.

A critical mass of nephrons is required to perform the kidney's daily tasks. Reductions in nephron number are associated with increased risk of hypertension and kidney disease.[1,2] The postnatal mammalian kidney lacks any clinically significant capacity to grow new nephrons to replace those that have been permanently lost.[3–8] The loss of nephron subunits may therefore be considered a principal cause of chronic kidney disease (CKD) and end-stage renal disease (ESRD).

Kidney transplantation, first successfully introduced over 60 years ago, remains the gold standard for treatment of ESRD. However, demand for transplantable kidneys far exceeds supply. In the United States, ~80% of patients that meet eligibility criteria for a kidney transplant do not receive one every year.[9] Among those fortunate enough to receive a kidney, a transplant is not a cure. Allograft recipients—even those receiving a so-called "perfect match"—must receive sustained immunosuppression for the lifetime of the transplant, which can have serious side effects. Acute rejection occurs within a year of transplant in ~10% of recipients, whereas the long-term graft longevity is ~12 years.[9–11]

Dialysis can be used as an alternative to kidney transplantation, and it does not require immunosuppression but it is associated with higher rates of mortality.[9] Dialysis machines clean the blood by equilibrating it with a balanced salt solution through a membrane, which does not substitute completely the complex filtration, reabsorption, secretion, and metabolic functions of kidney nephrons. Furthermore, dialysis is expensive, time-consuming and requires repeated access to the bloodstream, which can affect quality of life. In addition to kidney transplantation and dialysis, a limited number of pharmaceutical drugs are approved for use in individuals with kidney disease, including angiotensin-converting enzyme inhibitors, which reduce blood pressure and slow further deterioration of renal function, and corticosteroids, which reduce inflammation. However, these drugs are relatively few and primarily treat advanced stages of disease progression rather than root causes.[12]

The limitations of current therapies have inspired research and development of alternative approaches with the potential to prevent or treat kidney disease. Several emerging technologies have substantial promise to combat nephron loss, including stem cell therapy, gene therapy, and kidney regeneration. This chapter summarizes the state of the art in applying these strategies, both in the preclinical laboratory, and, where available, in the clinic.

KIDNEY STEM CELLS

A stem cell is an ancestor cell capable of producing a larger and more complex multicellular lineage. The process whereby stem cells change into more mature cell types is called differentiation. Alternatively, stem cells may duplicate themselves to produce two identical daughter cells, a process known as self-renewal. The balance between self-renewal and differentiation determines the fraction of stem cells versus mature cells in a tissue or organ. The ability of stem cells to generate many differentiated descendants makes them an attractive potential source of replacement tissue for the kidneys and other organs.

"Stem cell" is a general term that is not restricted to any specific biological entity or lineage.[13] A wide variety of unrelated cell types, with distinct origins and abilities, may be labeled stem cells. Various types of developmental stem cells arise transiently in the body at specific stages of embryonic development, proliferating and differentiating in a coordinated process to produce the bodily organs. In contrast, adult stem cells persist throughout postnatal life and contribute to the continuous renewal of rapidly growing organs such as blood, skin, hair, and intestine. Here, we summarize what is known about both developmental and adult stem cells for the kidney.

PLURIPOTENT STEM CELLS (ES AND IPS)

Pluripotent stem cells (PSC) can differentiate into all other cell types of the body, i.e., they have the potential for many different cell fates. PSC that arise during the natural development of an embryo are called embryonic stem (ES) cells. Vertebrate ES cells are a transient population that appears shortly after fertilization, and disappears after gastrulation, when the three embryonic germ layers have been established. ES cells can be cultured and expanded outside of the body in specialized growth media that sustain their self-renewal.[14–16] ES cells from different species have distinct cell culture requirements, morphological characteristics, and gene expression patterns. Mouse ES cells resemble the inner cell mass of the blastocyst-stage embryo, which is a tightly compacted ball of cells.[14,15] Human ES cells resemble the early embryonic epiblast, a disc-like, columnar epithelium that derives from the inner cell mass and is therefore further along in development.[16]

The nucleus of an adult cell can be reprogrammed (converted) to an ES cell-like state by forcibly altering its gene expression patterns. This can be accomplished by implanting the nucleus within an enucleated egg cell, a procedure known as somatic cell nuclear transfer (SCNT). The reconstructed cell resembles a fertilized egg (zygote) and can undergo embryonic development to produce an entire organism, which possesses genomic DNA identical to the donor's, and closely resembles the donor phenotypically. This process is called reproductive cloning.[17,18]

The first mammal to be successfully cloned from an adult cell was Dolly the sheep in 1996.[17] Reproductive cloning has since been successfully performed in a variety of species, including dogs and monkeys,[19,20] endangered animals,[21] and even from mice that have been deceased for decades.[22] However, cloning is highly inefficient, and successfully cloned animals frequently suffer from a variety of congenital maladies, which likely result from incomplete reprogramming of the somatic nucleus. These safety issues must be resolved before reproductive cloning can be ethically attempted in human beings.

In many species, including humans, ES cells can be derived in vitro from an SCNT blastocyst.[23,24] This process is called therapeutic cloning, because the resultant ES cells are expected to be highly immunocompatible with the donor, and could be used for cell therapy applications in that individual. However, therapeutic cloning remains an inefficient process and depends on the availability of human eggs, which are precious and can only be obtained from a female donor of reproductive age.[23,24] This has greatly limited the practicality of using SCNT for therapeutic purposes.

Induced pluripotent stem (iPS) cells are somatic cells that have been directly reprogrammed to an ES cell-like pluripotent state, without requiring eggs or SCNT. To generate iPS cells, a primary culture of mature, differentiated cells from any organ is forced to express a set of master regulatory genes that are normally expressed in ES cells.[25,26] When these cells are cultured under conditions that promote ES cell self-renewal, a small minority is "induced" (reprogrammed) into PSC.[25,26] Once reprogrammed, iPS cells are stable in the undifferentiated state and do not require forced expression of ES cell genes to sustain pluripotency.[25-29] iPS cells and ES cells are highly similar cell types, to the point that genome-wide gene expression analysis cannot easily distinguish between them.[30]

iPS cells can be utilized to produce a variety of differentiated cells, using protocols that work similarly for ES cells.[31–35] Compared with ES cells derived from SCNT, iPS cells are a much more convenient source of autologous PSC, because they can be generated using conventional cell culture techniques, and do not depend on the availability of human eggs. iPS cells have been generated from hundreds of patients.[36,37] With respect to immunocompatibility, the mitochondria of iPS cells are genetically identical to those of the original donor cell, unlike SCNT-derived ES cells, whose mitochondria derive from the host egg.[23,24]

Because iPS cells are derived directly from the original patient, they are predicted to be immunocompatible and would not require immunosuppression upon transplantation. Studies in mice support this prediction.[31,32] In a recent study in humans, a patient with wet (neovascular) macular degeneration was administered a graft of retinal pigmented epithelial cells derived from her own iPS cells.[38,39] After 25 months, the graft did not show any signs of rejection despite the complete absence of immunosuppressive medication.[38] Notably, a second patient from whom iPS cells were derived was not transplanted, due to concerns about potentially oncogenic somatic DNA mutations that arose in the iPS cells during the reprogramming process.[38] This highlights the importance of characterizing each new cell line to make sure that it is safe for transplant.

Collectively, ES cells and iPS cells are the two main types of PSC currently in use in the research laboratory and the clinic. PSC in culture represent a later stage of development than the zygote and cannot form a blastocyst on their own. In mice, however, PSC can be combined with extraembryonic stem cells to produce a blastocyst, which can develop into a whole, fertile mouse.[40,41] This demonstrates that PSC are truly pluripotent and can give rise to the entire body, including the kidneys. When cultured PSC are implanted into immunodeficient animals, they grow and differentiate into large tumors called teratomas, which contain tissues representing the three germ layers of the embryo.[16,25] A remarkable feature of PSC is that they can be cultured extensively in the undifferentiated state, for hundreds of passages, and still appear and function similar to earlier passages.[16,26]

Even after extensive passaging, PSC can be readily differentiated in vitro into a variety of somatic cell types, including cardiomyocytes, intestinal cells, cartilage, pancreatic cells, endothelial cells, and other lineages. Detailed regimens of specific growth factors are required to enrich PSC differentiation for specific cell types, which in some cases can form functional grafts in laboratory animals.[33–35] A number of PSC lines have now been generated from individuals with kidney disorders, including polycystic kidney disease (PKD), Alport syndrome, Wilms tumor, focal segmental glomerulosclerosis (FSGS), and systemic lupus erythematosus.[42–52] An immediate use of these cell lines has been to elucidate molecular and cellular mechanisms of human kidney disease, which was initially explored in nonrenal cell types.[42,52] Recently, it has also become possible to successfully differentiate PSC into the kidney lineage, as described later.

NEPHRON PROGENITOR CELLS AND ORGANOIDS

During embryonic development, ES cells differentiate into the three germ layers of the embryo, namely the endoderm, ectoderm, and mesoderm. The kidneys arise through an iterative series of reciprocal interactions between two stem cell populations, the ureteric bud (UB) and the metanephric mesenchyme (MM).[53,54] The UB differentiates into collecting ducts, while the MM differentiates into the tubules, podocytes, vasculature, and interstitial (stromal) cells.

Nephron progenitor cells (NPC) are a specialized subset of MM cells that differentiate into the podocytes, proximal tubules, and distal tubules of the nephron (Fig. 85.1A). In mammals, NPC are a transient population, like PSC. NPC express specific genes, such as SIX2, which balance self-renewal and differentiation during embryonic kidney development.[6,8] The final wave of differentiation is completed approximately at the time of birth, after which nephrogenesis ceases for the rest of the organism's life.[3–8]

In some fish species, a reservoir of NPC resides within a specialized niche even during adult life and becomes activated after kidney injury.[55] This differs from adult mammals, which lack NPC and the capacity to generate new nephrons. Under certain circumstances, however, NPC can persist in the postnatal mammalian kidney. An example of this is Wilms tumor, a pediatric renal cancer that contains structures that resemble developing kidneys, expresses NPC genes, and grows via a process that mimics nephrogenesis.[56–58] In transgenic mice, overexpression of the RNA-binding protein LIN28 in NPC results in the formation of Wilms tumors, which cause the kidneys to grow to massive proportions.[59] Using techniques such as these, it may be possible to sustain NPC postnatally, to augment kidney tissue. However, it is not clear yet whether the developmental program of nephrogenesis can be reactivated in adult cells, or how to control nephrogenesis in adult kidneys to produce healthy, functional kidney tissue, rather than tumors.

NPC have specialized requirements for survival and growth. For many years, long-term cultures of NPC could not be established in the laboratory. Recently, however, specific culture conditions have been identified that enable the stable propagation of mouse and human NPC in vitro, in some cases for over fifty passages, or approximately six months (Fig. 85.1B).[60–62]

Although there was no way to derive kidney cells from ES or iPS cells for many years, recent work by a number of groups has established methodologies to induce NPC differentiation from PSC.[50,63–65] Treatment of PSC with specific combinations of growth factors or chemical activators is sufficient to upregulate transcription factors associated with primitive streak mesendoderm, intermediate mesoderm, and NPC.[66,67] For human PSC, this involves activation of the canonical WNT signaling pathway. Optimization of these protocols has resulted in the successful differentiation of PSC into NPC.[50,63–65] Thus by reprogramming somatic cells to an iPS cell intermediate, it is now possible to differentiate and expand new NPC from any individual.

Kidney organoids are multicellular units in vitro containing structures that resemble nephrons, including tubular and podocyte components.[61,62] Kidney organoids are the products of NPC differentiation in vitro. Conversely, the ability to differentiate into kidney organoids is a litmus test for the presence of bona fide NPC. Kidney organoids in culture contain translucent, tubular structures that can be discerned by phase-contrast microscopy (see Fig. 85.1B).[50,60–65] At least three major nephron segments are present in kidney organoids: proximal tubules, distal tubules, and podocytes.[50,61–65] These segments form a continuum within the kidney organoid tubules and appear in a proximal-to-distal order that matches the nephron. Their appearance is a common denominator between different protocols and a defining characteristic of kidney organoids (Fig. 85.1C).[50,61–65]

Kidney organoids can be generated from primary and PSC-derived NPC, and they are similar with regards to marker expression and culture requirements.[50,60–65] Depending on the protocol used, organoids can range from smaller adherent structures enriched for nephron cell types (~200 μm diameter),[50] to unpurified aggregates containing both kidney and nonkidney cells (~3 mm diameter).[63–65] When injured with cisplatin or gentamicin, two well-characterized nephrotoxins, kidney organoids exhibit dose-dependent toxicity and express kidney injury molecule 1, a specific biomarker of acute kidney injury.[50,62,64,65] Tubular segments in kidney organoids furthermore absorb specific cargoes typical of kidney tubules.[50,65]

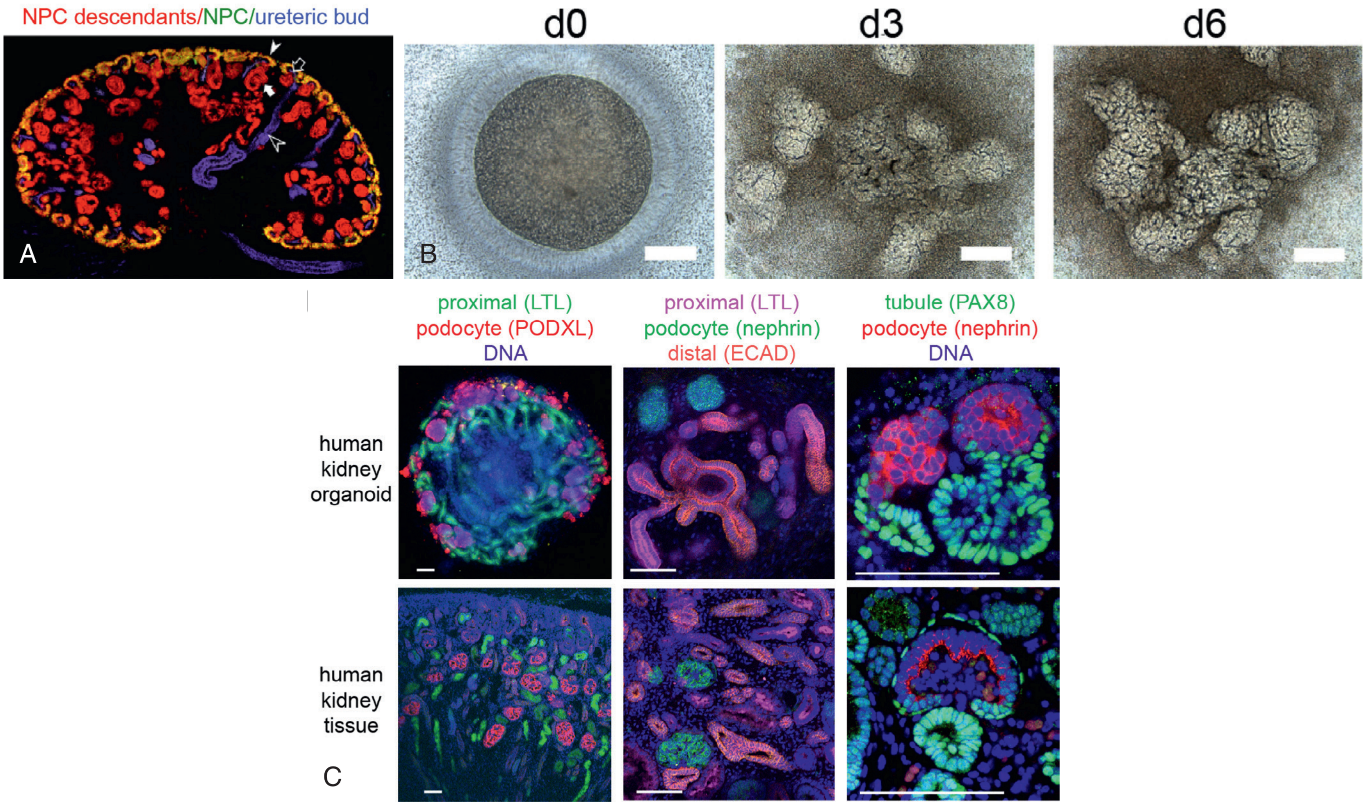

Fig. 85.1 Differentiation of mouse and human nephron progenitor cells (NPC) (A) Embryonic mouse kidney showing lineage tracing of NPC in differentiating nephrons. (From Kobayashi A, Valerius MT, Mugford JW, et al. Six2 defines and regulates a multipotent self-renewing nephron progenitor population throughout mammalian kidney development. *Cell Stem Cell* 2008;3:169–181.) (B) Nephron differentiation from human NPC cultures for 43 passages (~6 months). Scale bar, 200 µm. (From Li Z, Araoka T, Wu J, et al. 3D culture supports long-term expansion of mouse and human nephrogenic progenitors, *Cell Stem Cell.* 2016;19:516–529.) (C) Nephron segment profiles in human kidney organoids derived from PSC compared with developing kidneys. Scale bars, 100 µm. *LTL,* Lotus tetragonolobus lectin; *ECAD,* E-cadherin; *PODXL,* podocalyxin; *PAX8,* paired box 8. (From Freedman BS, Brooks CR, Lam AQ, et al. Modelling kidney disease with CRISPR-mutant kidney organoids derived from human pluripotent epiblast spheroids. *Nat Commun* 2015;6:8715; Cruz NM, Song X, Czerniecki SM, et al. Organoid cystogenesis reveals a critical role of microenvironment in human polycystic kidney disease. *Nat Mater.* 2017;16:1112–1119; and Kim YK, Refaeli I, Brooks CR, et al. Gene-edited human kidney organoids reveal mechanisms of disease in podocyte development. *Stem Cells* 2017;35(12):2366–2378.)

Importantly, these biomarker and transporter characteristics differ from those of undifferentiated PSC epithelia (epiblast), indicating that they are lineage-specific.[50] Most impressively, kidney organoids can recapitulate three-dimensional phenotypes of genetic kidney disease, including polycystic kidney disease and podocyte disorders.[50,68,69] The organoids therefore mimic essential properties of kidney tissue in vivo.

Kidney organoids in their current form also have limitations. Although several groups have shown that kidney organoids contain podocytes, proximal tubules, and distal tubules, all of which derive from NPC,[50,61–65] there are differences in opinion over whether these cultures also contain UB and a collecting duct network, which would represent a major advance over MM derivatives alone.[65,70] Complicating this question, PSC from different individuals exhibit substantially different abilities to form kidney organoids, depending on the protocol used for differentiation.[68] This variability, which is not unusual for PSC derivatives, poses a significant technical challenge for using these techniques for patient-specific kidney regeneration and disease modeling. There are also concerns regarding the maturity of the epithelial cells within kidney organoids. There is no source of nephron perfusion, and the tubules do not form expansive lumens or brush borders.[50,64] Podocytes do not form bona fide foot processes with secondary and tertiary interdigitations and appear to be arrested at the capillary loop stage of glomerular development.[50,69,71] Development of the mammalian kidney proceeds via a series of successive phases: the pronephros, mesonephros, and metanephros. The pronephros is the most primitive form of kidney, while the metanephros is the most developed. The metanephros forms the definitive adult kidney. It is not yet clear whether organoids derived from PSC represent pronephric, mesonephric, or metanephric kidneys.

In addition to kidney epithelial lineages, other cell types also arise within kidney organoid cultures. For primary NPC, the identity of these nonepithelial cell types is not yet clear.[60–62] For PSC-derived organoid cultures, several nonepithelial cell types have been identified, including endothelial cells, fibroblast-like stromal cells, and neurons.[50,65] Contamination with nonkidney cell types is to be expected given that these protocols begin with PSC, which can naturally differentiate into any cell type. The presence of these nontubular cell types could be advantageous; for instance it has been observed that endothelial cells within these cultures interact with kidney tubular cell types and podocytes.[50,65] However, the presence of nonkidney cells also has disadvantages, in that it decreases the purity and yield of the resultant kidney structures, which cannot be easily separated from the contaminants.

One promising strategy that remains to be fully developed is to directly reprogram patient cells into NPC or UB, rather than into PSC. Direct reprogramming approaches have succeeded to a varied degree for other cell types, such as cardiomyocytes and neurons.[72,73] As culture conditions for culturing NPC are now established, and many of the major transcription factors are known, the field is primed for such an advance. Efforts in this direction have succeeded in inducing changes to cells that partially resemble NPC or kidney tubules, although sustained expression of the transgenes was required to maintain the phenotype, and no kidney organoids were generated, indicating that reprogramming was insufficient to induce bona fide NPC.[74,75] Achieving the goal of directly reprogramming somatic cells into NPC could be an important step in establishing a more reproducible and practical strategy for kidney cell therapy. A limitation of this approach, however, is that NPC or UB have a more limited potential than PSC and cannot generate entire kidneys on their own.

ADULT STEM CELLS

Adult stem cells are specialized resident cells that continuously repopulate a specific organ throughout life. They typically reside within a specialized structure, or niche, within the tissue. Examples of adult stem cells are intestinal stem cells, epidermal stem cells, and muscle satellite cells. Adult stem cells are typically found in tissues or organs that have a high turnover rate at homeostasis. They first arise during fetal development, and persist into old age, for the lifetime of the organism.

Although mammalian kidneys are incapable of wholly regenerating new nephrons, they do possess the ability to replace cells within existing nephrons. When subjected to acute tubular damage, such as after ischemia-reperfusion injury, new tubular epithelial cells arise within the kidney to heal the wounded epithelium.[76,77] Evidence from mice carrying specific lineage reporters indicates that the new epithelial cells derive from older epithelial cells, which appear to dedifferentiate and repopulate the basement membrane in a segment-specific manner.[76–79]

Subsets of epithelial cells can exhibit an increased propensity to proliferate and repopulate injured tubules.[77,80] Nevertheless, in contrast to classic adult stem cells in tissues such as skin, blood, and intestine, no specialized stem cell niche can be readily discerned in the mature, quiescent epithelium of the kidney nephron. Furthermore, this epithelium does not undergo rapid and constant self-renewal under conditions of homeostasis. For these reasons, kidney tubular epithelial cells do not completely represent a classic adult stem cell. Rather, they may more closely resemble the beta cells of the pancreas, which grow and maintain their number via a process of self-duplication of other terminally differentiated beta cells.[81]

Unlike tubular epithelial cells, podocytes in the adult kidney permanently exit the cell cycle and are incapable of repopulating their own epithelium after injury.[82] Rather, the primary response of podocytes to injury is hypertrophy, an increase in the size of individual cells.[83] However, recent work suggests that podocytes increase in number by up to 20% during human adolescence, suggesting a progenitor source.[84] Furthermore, patients with kidney disease shed podocytes into their urine at a faster rate than would be expected based on clinical progression, which suggests that the body may harbor an innate capacity to replace lost podocytes.[85,86] Lineage tracing experiments in animals suggest that the new podocyte-like cells do not derive from other podocytes.[87–89]

One possible reservoir of replacement cells resides within the population of parietal epithelial cells (PEC) that lines Bowman's capsule. A subpopulation of cells in this compartment coexpress podocyte and PEC markers, and has been referred to as "glomerular epithelial transitional cells" or "parietal podocytes" for their proposed ability to transdifferentiate into podocytes during adolescence or after injury.[86,88–93] However, there is also evidence that podocytes can transition into the parietal compartment in injury states, therefore the role of the transitional podocytes in glomerular repair has not yet been settled.[94] A second possible reservoir of replacement cells for podocytes are cells of renin lineage (CoRL), which are vascular smooth muscle cells in the juxtaglomerular compartments which produce renin.[95] Like parietal podocytes, CoRL can migrate into the glomerulus and adopt features of podocytes, including the expression of podocyte-specific markers and the elaboration of foot processes.[96,97]

Nonrenal adult stem cells have also been investigated for their potential to contribute to kidney repair after injury. Hematopoietic stem cells (HSC) naturally reside in the bone marrow and have the ability to repopulate the entire blood lineage. HSC can also be found in other sites, such as amniotic fluid and umbilical cord blood. Initially, it was surmised that circulating HSC might also be capable of differentiating into kidney cells. However, a series of studies has now established that HSC do not differentiate into tubular epithelial cells after injury.[98,99]

Mesenchymal stem cells (MSC), also called multipotent stromal cells, are populations of interstitial cells with expansive proliferative potential. A hallmark of MSC is that they can be induced to differentiate in vitro into fat, cartilage, and bone. MSC were first identified in the bone marrow but are now recognized to reside within a variety of different organs. The kidney contains a reservoir of MSC in the peritubular interstitium and vasculature, which enact fibrotic scarring after injury.[100,101] Human MSC can be similarly purified from the renal artery.[102] Ablation of MSC from mouse kidneys induces acute tubular necrosis and capillary rarefaction but can also be protective from vascular calcification in chronic disease.[103,104]

KIDNEY REGENERATION

Transplantation experiments in several organs, including the heart, liver, and pancreas, suggest that somatic cells and tissues derived from stem cells can engraft and function in animal models.[33–35] Whether such approaches might also be efficacious to treat kidney disease is currently being investigated in experimental models, which are optimizing the delivery of kidney stem cells and testing their potential for clinical kidney regeneration.

STEM CELL THERAPY

Cell therapy involves the direct administration of cells into the body for healing purposes. The units of therapy in this approach are single cells. For regenerative medicine, the ultimate objective of cell therapy is to establish a long-term graft with the capacity to perform organ functions. A practical example is bone marrow transplantation, in which HSC are the units of therapy, engraft in the bone marrow, and repopulate the entire blood lineage.[105]

Intravenous administration describes the direct injection of dissociated cells into the bloodstream using a syringe. It is the simplest delivery route for cell therapies and is used for HSC therapy in the clinic. Kidney cells, however, are different from blood cells and do not typically circulate throughout the body. The kidney is furthermore a densely-packed organ with no obvious route for stem cells to traverse from the bloodstream into the nephrons. Whether kidney stem cells have the ability to engraft and regenerate the kidney after intravenous administration therefore needs to be tested in preclinical animal models. In these experiments, the kidneys are typically subjected to acute injury. This damages the glomerular filtration barrier, which can enhance penetration of cells into the kidney and subsequent engraftment.

In one example, human iPS cell-derived cells expressing a variety of NPC and adult kidney cell markers were injected into the mouse tail vein 24 hours after administration of the nephrotoxic drug cisplatin.[106] Extensive engraftment was reported in proximal tubules, which coincided with a 55% reduction in urea levels in treated mice, compared with control animals administered with saline or undifferentiated iPS cells.[106] These experiments suggest a possible benefit of iPS-derived kidney cells on kidney injury. However, the isolated cells were not shown to demonstrate the ability to form kidney organoids with segmented nephrons. It is therefore unclear whether the implanted cells contained bona fide NPC or whether new nephrons were actually formed.

Intravenous administration has also been applied to adult kidney cell populations. Human glomerular epithelial transitional cells (see earlier), administered intravenously into a mouse model of chemically-induced podocytopathy, were found in glomeruli, and were associated with a decrease in proteinuria.[107] These cells also contributed to tubules after acute injury.[80] As these cells cannot form new nephrons, this approach seeks to repair and replace, rather than to completely regenerate.

MSC can be readily obtained, for instance from a patient's adipose tissue. Intravenous administration of MSC in experimental models can have a beneficial effect on ischemia-reperfusion injury.[99,102,108] This benefit can be obtained even in the absence of MSC engraftment, likely via a paracrine effect. However, MSC administered to injured kidneys do not contribute tangibly to new nephron formation and can differentiate ectopically into undesirable fat cells or fibroblasts within glomeruli.[108,109] Collectively, these findings suggest that intravenous administration of cell therapeutics may provide some benefit in cases where the glomerular filtration barrier has been compromised but may also have unwanted side effects.

In the technique of parenchymal injection, cell therapies are directly delivered into the kidney cortex using a syringe. In one study, 5×10^5 primary NPC were purified from human fetal kidneys, and injected directly into kidney parenchyma of immunodeficient mice subjected to 5/6 nephrectomy.[110] Follow-up revealed engraftment of the NPC, which was associated with a modest improvement in serum creatinine.[110] In another instance, newborn mouse kidneys were implanted with dissociated cultures of kidney organoids derived from human ES cells.[50] Three weeks later, human proximal tubular structures were observed alongside mouse proximal tubules within the cortex.[50] In a similar model, mouse primary NPC injected into newborn mouse kidneys formed chimeric tubules and glomeruli with the host mouse, which could be distinguished based upon expression of a fluorescent tracer (Fig. 85.2A).[62]

Parenchymal injection has the advantage of placing the cells in direct physical contact with other nephrons. However, the kidney parenchyma is tightly packed with blood vessels and nephrons. This limits the ability to introduce large numbers of stem cells and target them to specific locations. In addition, injection of cells is likely to damage existing kidney tissues. Illustrating this, human iPS cells were differentiated into cells expressing the NPC markers OSR1 and SIX2, treated with growth factors, and injected into the kidney parenchyma immediately after induction of ischemia-reperfusion injury.[111] Although some engraftment was observed, the authors noted difficulty administering a large number of cells directly into the parenchyma, and were concerned about causing damage to the kidneys using this method.[111]

Subcapsular implantation describes insertion of cells in between the kidney cortex and the renal capsule, a thin, fibrous membrane surrounding the kidneys.[111] This space can accommodate a large number of cells. Subcapsular implantation of stem cells into the mouse kidney has been performed by several groups.[63,66,111] Using this method, the foreign growth can be distinguished from existing kidney structures, from which it is largely separated geographically, and analyzed side-by-side.

Using protocols that produce kidney organoids, mouse or human NPC were implanted beneath the capsule of immunodeficient mice.[63,71] These growths contained a variety of stromal and epithelial structures. Some of these resembled primitive nephrons, including both tubules and glomerulus-like rosettes, both of which appeared more diffuse and eosinophilic than the neighboring kidney tissue (Fig. 85.2B).[63,71] These growths also contained endothelial structures, derived from the mouse host.[63,71] Notably, most of the growth area appeared stromal. Similarly, subcapsular implantation of kidney-like cells derived from human ES cells resulted in large growths that expressed AQP1 at similar levels to neighboring kidney cortical cells.[66] However, these growths did not morphologically resemble the neighboring parenchyma histologically and appeared instead to contain diffusely staining stroma instead of tubular structures.[66]

In a functional test, NPC-like cells derived from human iPS cells subjected to subcapsular implantation appeared to significantly ameliorate the effects of kidney ischemia-reperfusion injury prior to the transplant, as determined by serum creatinine and urea levels as well as by histological injury scores, compared with mice receiving grafts of undifferentiated iPS cells or mock injections.[111] Because the capsule is a distinct location from the rest of the kidney, with no

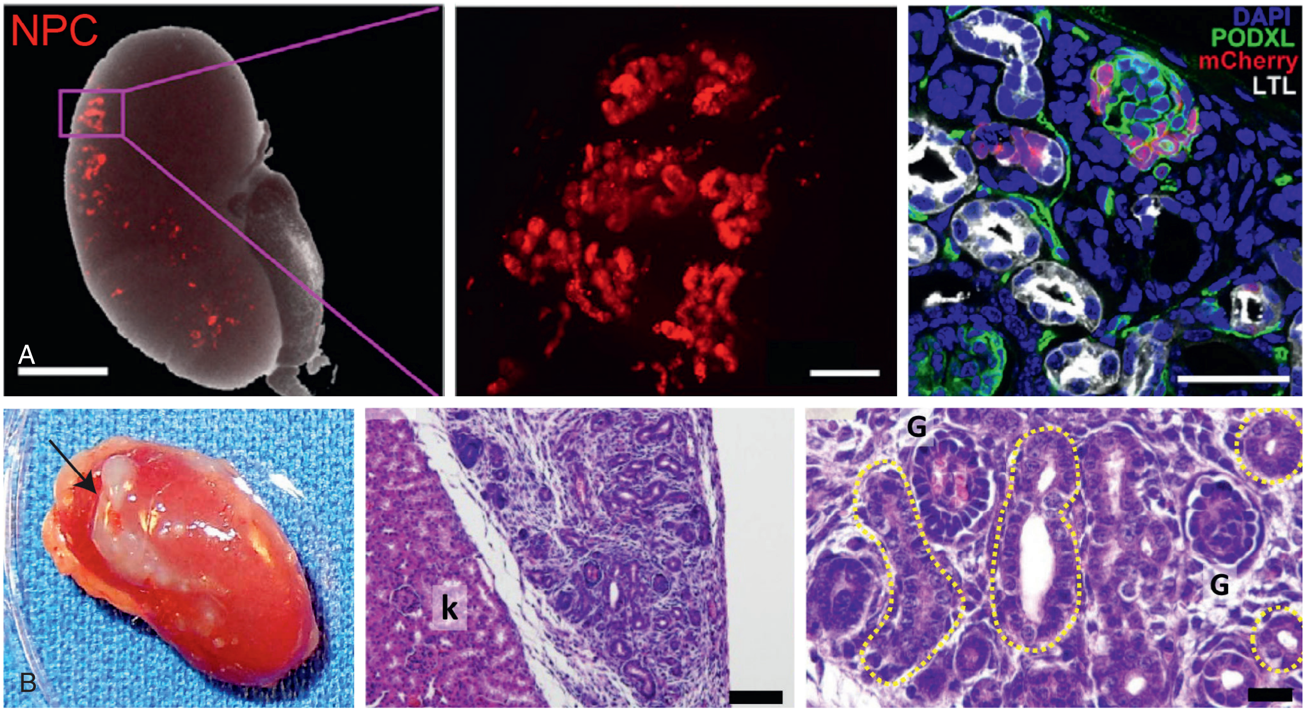

Fig. 85.2 Delivery of kidney cell therapies (A) Eight-day-old mouse kidney 1 week after cortical injection of mCherry-labeled nephron progenitor cells (NPC; *red*). Chimeric nephrons are observed by immunofluorescence. Scale bars are 1 mm, 100 μm, and 50 μm *(from left)*. (From Li Z, Araoka T, Wu J, et al. 3D culture supports long-term expansion of mouse and human nephrogenic progenitors, *Cell Stem Cell.* 2016;19:516–529). (B) Growths of human NPC derived from pluripotent stem cells (PSC) and implanted beneath the mouse kidney capsule. *Arrow* indicates NPC-derived mass. Nascent tubules are outlined in histology images. *k,* Host kidney; *G,* glomerulus-like structures. Scale bars, 100 μm (low magnification) and 20 μm (higher magnification). (From Lam AQ, Freedman BS, Morizane R, et al. Rapid and efficient differentiation of human pluripotent stem cells into intermediate mesoderm that forms tubules expressing kidney proximal tubular markers. *J Am Soc Nephrol.* 2014;25:1211–1225, and Taguchi A, Kaku Y, Ohmori T, et al. Redefining the in vivo origin of metanephric nephron progenitors enables generation of complex kidney structures from pluripotent stem cells. *Cell Stem Cell* 2014;14:53–67.)

obvious way to connect with the kidney, this functional improvement is hypothesized to be a paracrine effect.[111] The NPC-like cells expressed a variety of paracrine factors such as VEGF and HGF that might promote functional recovery.[111] A limitation of this study is that the iPS-derived cells were not shown to form kidney organoids, therefore it is unclear whether they were true NPC.

Because kidney disease has systemic causes and effects, implantation of nonrenal cells may also be therapeutically beneficial. For instance, subcapsular implantation of erythropoietin-expressing liver cells derived from iPS cells improves renal anemia in a mouse model.[112] Similarly, implantation of human beta cells derived from PSC, optimized to secrete insulin in response to glucose, rapidly reduced hyperglycemia in a small cohort of diabetic mice.[113] Whether the kidney capsule will provide a suitable microenvironment for engraftment of nonrenal cells in human beings is not yet clear.

To summarize, based on the studies mentioned earlier, implantation of NPC can result in the differentiation of new nephron-like structures in mammalian kidneys in vivo. However, tissues derived from kidney cell therapies in their present form appear disorganized and immature, and it is unclear whether they can integrate into the extant collecting duct or vascular networks. Only in a limited number of these cell therapy studies has any beneficial effect been shown in animal models of kidney disease.[106,107,110,111] Safety must also be considered, as immature kidney cells or contaminating nonkidney cells could be harmful or tumorigenic. This is illustrated by recent cases in which administration of autologous adipose stem cells caused deterioration instead of improvement in the kidneys or the eyes.[109,114] Further preclinical studies in animal models is therefore required to establish therapeutic proof of principle, compared with the gold standard of kidney transplant.

Clinical Relevance

Over 700 clinics across the United States offer stem cell therapies that are unproven and have not been FDA approved. Treatments typically involve intravenous or intramuscular administration of nonrenal adult stem cells. These therapies are being advertised for an assortment of maladies, including kidney disease. Patients should be reminded that these unproven stem cell treatments are associated with significant risks, including damage to glomeruli, unexpected immune responses, and tumorigenesis, and may compromise insurance claims. In many cases, the physicians involved in these practices may be under investigation by the local or state medical board.

BIOARTIFICIAL KIDNEYS

Biomedical engineering or bioengineering is the construction of artificial tissues and organs. This may include, for instance, the combination of natural and synthetic materials to create a patterned scaffold onto which cells are layered in specific arrangements. Bioengineering is typically performed outside of the body, in a lab dish or on a silicon chip. Strategies are currently being developed to generate artificial devices with kidney-like functionalities.

One approach is to utilize existing organs as scaffolds for cells. Whole kidneys are perfused with a detergent agent, which decellularizes the organ, leaving only the extracellular matrix intact in its original shape (Fig. 85.3A). Cells are then reintroduced into the kidney to recellularize it, either through the vasculature (anterograde) or from the ureter (retrograde). Decellularized rodent kidneys can be partially recellularized from primary kidney cells or PSC.[115,116] However, there is no clear way to target specific cell types to designated segments within the nephron, and it is unclear whether recellularization confers upon the scaffold the ability to function like a kidney, or filter urine.[117,118]

A "kidney on a chip" device consists of a synthetic scaffold seeded with kidney cells and subjected to continuous perfusion from a reservoir, which is attached to the device via tubing.[119–121] The scaffold itself may be molded from a synthetic material, naturally occurring extracellular matrix, or a combination of these. These devices tend to be small, about the size of a credit card, and are limited to specific segments of the nephron, such as the proximal tubule (Fig. 85.3B). Toxicity and transport assays indicate that these "three-dimensional" cell culture devices recapitulate nephron functions more accurately than monolayer cultures.[119–121] Currently, the primary application of these systems is to model injury responses in human kidney cells.

To bioengineer larger structures appropriate for transplantation, an emerging technology is 3D bioprinting. This approach utilizes robotic automation to seed cells in patterned layers of biomaterials, resulting in the manufacture of macroscopic, complex tissue constructs.[122–124] To maintain structural integrity, the cells are embedded in biodegradable (sacrificial) hydrogels of sufficient stiffness to withstand transplantation. The mold for bioprinted constructs can be based on a digitalized image, and microchannels can be included to enhance diffusion and cell survival.[122] This technique has primarily been applied to structural and connective tissues, which are relatively simple, and nevertheless remain less mature or organized than the corresponding tissues in vivo. Moreover, bioprinted tissue constructs have not been demonstrated to exhibit long-term functionality. Due to the intricate nephron infrastructure of the kidney, 3D bioprinting of this organ has not yet been attempted and represents an important frontier for this line of research.

Artificial kidneys are medical devices capable of performing renal replacement therapy. Conventional artificial kidneys (dialysis machines) are limited to intense periods of therapy, interspersed with periods of nonuse. In contrast, a wearable artificial kidney (WAK) is a miniaturized, mobile machine that continuously dialyzes the blood, based on dialysate-regenerating sorbent technology (Fig. 85.3C).[125,126] A recent WAK trial suggested that the device could achieve effective uremic solute clearance over a 24-hour period, and no serious adverse events were observed. Patients reported greater satisfaction with the WAK, compared with conventional dialysis, but technical difficulties and performance variabilities were also observed.[126]

A bioartificial kidney, also known as a renal assist device (RAD), consists of a hollow tubular scaffold lined with kidney tubular epithelial cells (similar to a kidney-on-a-chip device) that is designed to simulate the reabsorption and secretion functions of the nephron tubule. The epithelial cells perform vectorial transport of nutrients or toxins between the luminal (inner) and peritubular (outer) filtrates, which are confined to separate compartments. The luminal filtrate can then be discarded or recycled, while the peritubular filtrate is returned to the patient, enriched with nutrients and cleansed of uremic solutes.

A prototype RAD, consisting of approximately one billion porcine kidney cells grown to confluence along the inside of the hollow fibers of a conventional hemodialysis filter cartridge, was effective in curtailing acute uremia in a canine model.[127] The device also provided kidney metabolic functions, which are not provided by conventional dialysis. Subsequently, a similar device lined with human cells was investigated in clinical trials, which suggested both safety and efficacy.[128,129] However, a follow-up phase II study was not completed, and further work is necessary to reproduce these findings.

An ambitious proposal is to make bioartificial kidneys implantable in the body (Fig. 85.3D).[130,131] It is envisioned that such a device could use specialized hemofiltration membranes that have been optimized to mimic glomerular slit diaphragms.[132,133] The filtrate from these membranes would pass along to a bioreactor that would be sustained inside the body. Major technical and conceptual challenges for this approach include: 1. Keeping the cells inside the device alive, protected from the patient's immune system, and in an intact tubular conformation that supports function; 2. Achieving suitable pump power and sufficient water supply, as extracorporeal dialysis can use up to 200 L of water over four hours, whereas water in the implanted bioartificial kidney would be restricted to that which can be supplied by drinking; 3. Designing a nephron-like device of sufficient complexity and organization to properly balance secretion, reabsorption, and concentration functions, resulting in hemofiltrates of appropriate composition and volume. Overcoming these challenges could produce a radically different form of renal replacement from conventional therapies.

Clinical Relevance

Efforts to generate WAK devices began in the 1970s, but have recently accelerated, due in part to advances in miniaturization. The new generation of wearable or implantable artificial kidneys is envisioned as the renal equivalent of the cardiac pacemaker or pancreatic insulin pump. The continuous operation of the device is predicted to have a stabilizing effect on blood volume and composition, while its mobility affords patients with a greater sense of agency, compared with conventional hemodialysis. However, these devices remain in the prototype form, and require further optimization and testing before they are fit for regular use in patients.

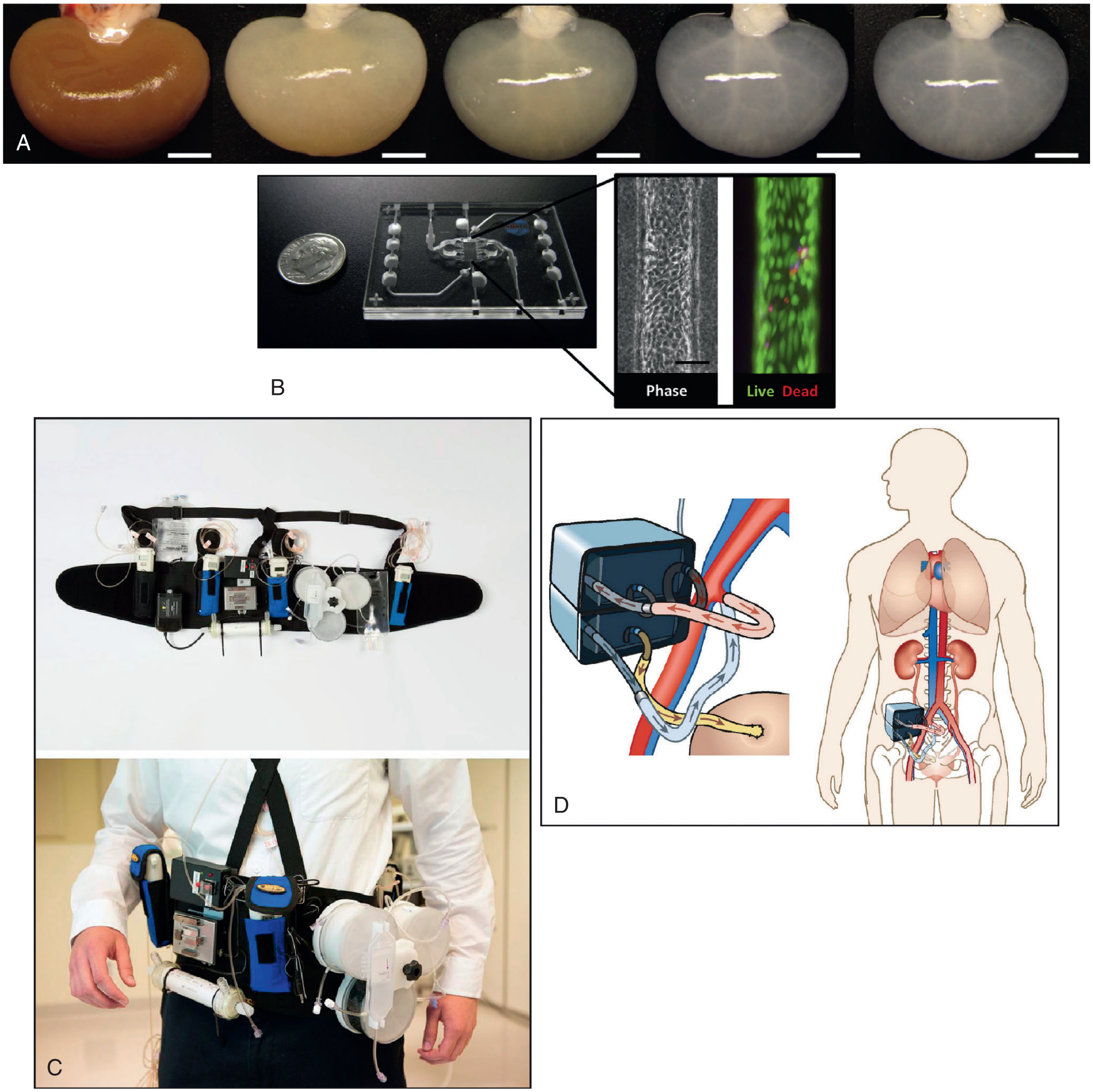

Fig. 85.3 Kidney bioengineering (A) Rat kidney subjected to progressive decellularization. Scale bars, 5 mm. (From Caralt M, Uzarski JS, Iacob S, et al. Optimization and critical evaluation of decellularization strategies to develop renal extracellular matrix scaffolds as biological templates for organ engineering and transplantation. *Am J Transplant.* 2015;15:64–75.) (B) Kidney-on-a-chip device, with dime for scale. Inset shows central tubule lined with kidney tubular epithelial cells. Scale bar, 50 μm. (From Weber EJ, Chapron A, Chapron BD, et al. Development of a microphysiological model of human kidney proximal tubule function. *Kidney Int.* 2016;90:627–637.) (C) Prototype wearable artificial kidney tested in patients. (From Gura V, Rivara MB, Bieber S, et al. A wearable artificial kidney for patients with end-stage renal disease. *JCI Insight.* 2016;1(8):e86397.) (D) Theoretical schematic of implantable bioartificial kidney, using iliac vessels for blood supply, and with ultrafiltrate draining to the bladder. (From Fissell WH, Roy S, Davenport A. Achieving more frequent and longer dialysis for the majority: wearable dialysis and implantable artificial kidney devices. *Kidney Int.* 2013;84:256–264.)

METANEPHRIC IMPLANTS

A limitation of cell therapy and bioengineering approaches is that they involve relatively simple cell populations and configurations, which do not recapitulate the intricate architecture of the kidney. The goal of metanephros implantation is to grow whole kidney organs in situ inside living hosts. The metanephros first appears during the fifth week of human development and contains NPC, UB, vascular cells, and stromal cells organized within a growing, branching cortex (Fig. 85.4A).[7,134] When multiple metanephroi (also called kidney analgens, rudiments, or primordia) are implanted into adult hosts, they can grow and differentiate into kidney-like masses.[135–139]

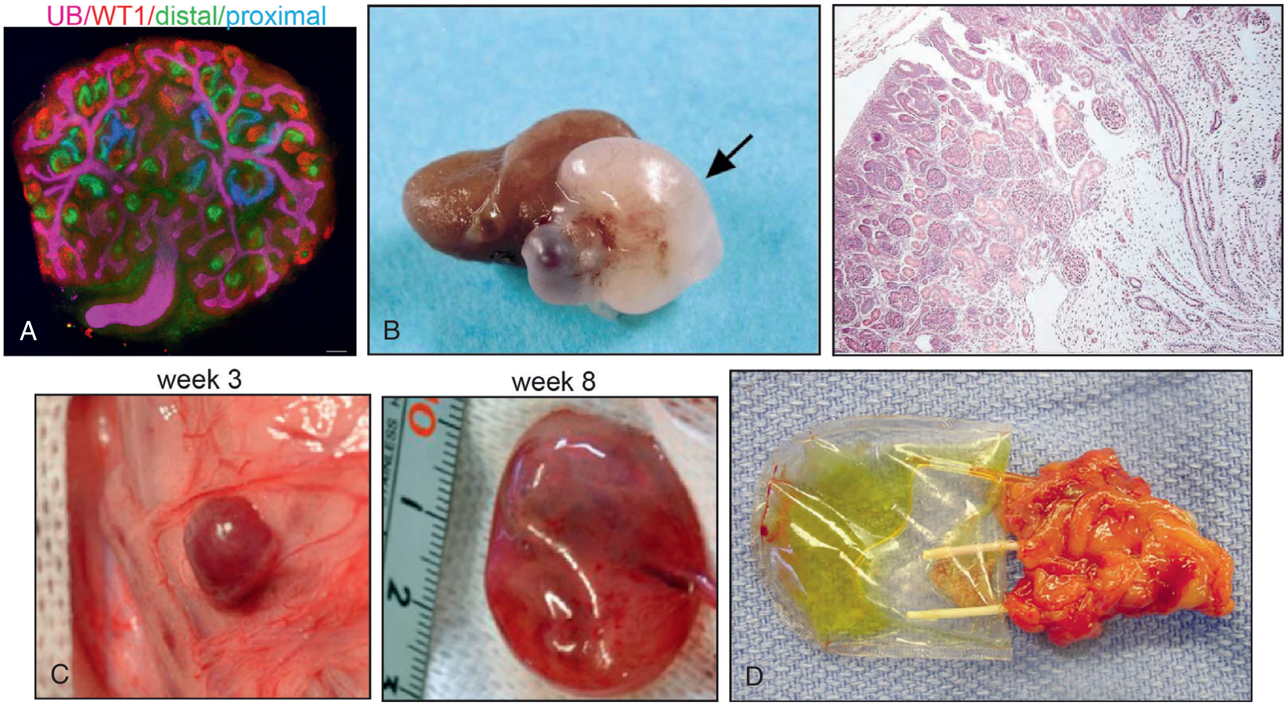

Fig. 85.4 Ectopic growth of metanephroi. (A) Mouse metanephros cultured in vitro, delineating nephron lineages naturally present in these structures. *UB,* Ureteric bud; *WT1,* Wilms tumor 1. Scale bar, 100 μm. (From Barak H, Boyle SC: Organ culture and immunostaining of mouse embryonic kidneys. *Cold Spring Harb Protoc.* 2011;2011:pdb prot5558.) (B) Human growth *(arrow)* arising from metanephric tissues implanted beneath the kidney capsule of an immunodeficient mouse, 8 weeks after transplant. Histology of the growth is shown at right. (From Dekel B, Burakova T, Arditti FD, et al. Y: Human and porcine early kidney precursors as a new source for transplantation. *Nat Med.* 2003;9:53–60.) (C) Pig metanephroi implanted in the omentum of a syngeneic host, 3 weeks *(left)* and 8 weeks *(right)* posttransplant. Hydronephrosis is observed at later time points. (From Yokote S, Matsunari H, Iwai S, et al. Urine excretion strategy for stem cell-generated embryonic kidneys. *Proc Natl Acad Sci U S A.* 2015;112:12980–12985.) (D) Bioengineered renal construct seeded with the metanephric cells of a cloned cow, retrieved from the SCNT donor 12 weeks after histocompatible transplant. (From Lanza RP, Chung HY, Yoo JJ, et al. Generation of histocompatible tissues using nuclear transplantation. *Nat Biotechnol.* 2002;20:689–696.)

Metanephric implants can grow in a variety of bodily locations, including beneath the kidney capsule, the anterior chamber of the eye, and the omentum. Implantation of allogeneic rat metanephroi beneath the kidney capsule or into the anterior chamber of the eye resulted in the rapid vascularization of the grafts within 10 days.[135] The grafts, which contained metanephric mesenchyme and ureteric buds when implanted, differentiated extensively into nephron structures, which accumulated intravenously-injected IgG. However, the rapidly-growing allografts also provoked a strong immune response and were rejected after 2 weeks.[135] Matching of rat recipient and host strain allowed implanted metanephroi to survive for up to four weeks. The resultant metanephric implant grew to several times its original size, although the graft never approached the size of an actual adult kidney.[136]

Metanephroi grown in the omentum allow for efficient anastomosis of the metanephric ureter to the ureter of a nephrectomized kidney of the host animal (ureteroureterostomy).[136,137] When the contralateral kidney was subsequently removed, urine production and prolongation of life was observed in the recipient rats, although the effects were relatively modest and insufficient to replace kidney function.[136,137] Although this model involved matching of rat strains, an immune response was still observed, and tolerance was limited to metanephroi whereas adult kidneys transplanted in a similar way were rapidly rejected.[136]

Immunodeficient mice have been used as graft recipients to study the formation of metanephric implants from other species, including pigs and humans.[138] Graft success was dependent on gestational age, with younger grafts (embryonic week 7) growing much more successfully compared with older metanephric tissues or adult kidneys.[138] Under these conditions, and over approximately 8 weeks of growth, both human and porcine metanephroi grew to large sizes in the mouse kidney capsule, forming grafts that rivaled the size of the attached mouse kidneys and that included regions of well-formed cortex (Fig. 85.4B).[138]

Even if the metanephric implant grows to an appropriate size, it is important to provide it with conditions where it might be able to function properly. This is necessary both to confer the desired functional benefit to the host, and also to ensure that the graft develops without disease. Comparison of porcine metanephric implants with and without the associated cloaca (bladder primordial) revealed that inclusion of a bladder component was important for establishing a drain for the implant (Fig. 85.4C).[139] In the absence of cloaca, the implants grew but showed signs of hydronephrosis, whereas inclusion of the cloaca reduced tubular dilation and fibrosis, and increased both the quantity and concentration of uremic solutes within the urine produced by the grafts.[139] Anastomosis of the graft ureter to the host ureter also resulted in improved graft health in both rats and pigs, and prolonged survival of anephric rats.[139]

Implantation of metanephroi can also be combined with bioengineering and stem cell approaches to produce immunocompatible tissue constructs. To investigate such a strategy, the metanephros of a 56-day-old bovine calf embryo produced by reproductive cloning was harvested, seeded onto a scaffold of polycarbonate and collagen, and reimplanted back into the original donor cow. The resultant renal constructs secreted yellow, "urine-like" fluid containing urea nitrogen and creatinine, which accumulated in an attached reservoir over the next twelve weeks (Fig. 85.4D).[140] Thus there is significant potential for combining iPS cells, metanephroi, and bioartificial kidney technologies.

For such approaches to become a realistic option, a reliable source of metanephroi is needed. Human metanephroi appear early in development and cannot be obtained for transplantation. Older kidneys may be less suitable for transplantation, compared with metanephroi.[136,138] Therefore, the possibility of using metanephroi from other species, which can be farmed, is being investigated.

XENOTRANSPLANTS AND CHIMERAS

Xenotransplantation refers to implantation of grafts from a different species than the recipient. The ability to harvest organs from other species, if successful, would present a possible solution to the current shortfall of donor organs. In early clinical trials of xenotransplantation in the 1960s, chimpanzees were used as donors but the recipients did not survive beyond 2 months. Subsequently, pigs have emerged as the leading potential donor species for such a procedure because they are domesticated, inexpensive, readily available, and their kidneys resemble human kidneys in size, architecture, and physiology.[141,142]

Historically, xenotransplantation approaches have been stymied by safety issues. First, the body's immune system will rapidly reject cells from other species, due to an abundance of nonself antigens. Xenograft rejection is therefore more rapid and severe than allograft rejection. Pig cells are particularly vulnerable, because they express specific carbohydrate antigens such as galactosyl-α1,3,-galactose (Gal) that provoke hyperacute (within minutes) and acute (within days) rejection responses in humans and other primates, which can cause graft failure shortly after transplant (Fig. 85.5A).[143,144]

Progress has been made toward mitigating these acute rejection responses. Miniature pigs lacking Gal have been generated, by genetic engineering in a cultured cell line followed by reproductive cloning to produce the animals.[145] Kidneys from these minipigs provoke a reduced hyperacute rejection response, but are nevertheless rejected in the weeks following transplant.[143,146] Combination of Gal-deficient kidneys with nonconventional immunosuppression regiments that block the costimulatory complement cascade further increases the maximum survival of porcine renal xenotransplants to greater than seven months.[147,148] These experiments have been performed in nonhuman primates, but have not yet been attempted in humans.

An alternative approach that has been proposed is to grow human organs in animals. A technique called interspecies blastocyst complementation (IBC) has recently been developed, as a step towards this goal.[149,150] In IBC, PSC from a donor species are implanted into blastocyst-stage embryos of a recipient species deficient in their ability to grow a specific organ. As a result, the donor cells "complement" the deficiency and form an organ that is largely donor-derived in an otherwise foreign host. Because the resultant animal comprises a mixture of host and donor tissues, it is called a chimera. IBC is still in its infancy and has not yet been successfully used with human PSC. In proof of principle experiments, however, researchers have grown a variety of rat organs in mice, including pancreas, heart, and eyes, which grow to the size of the recipient's organ (Fig. 85.5B).[149,150]

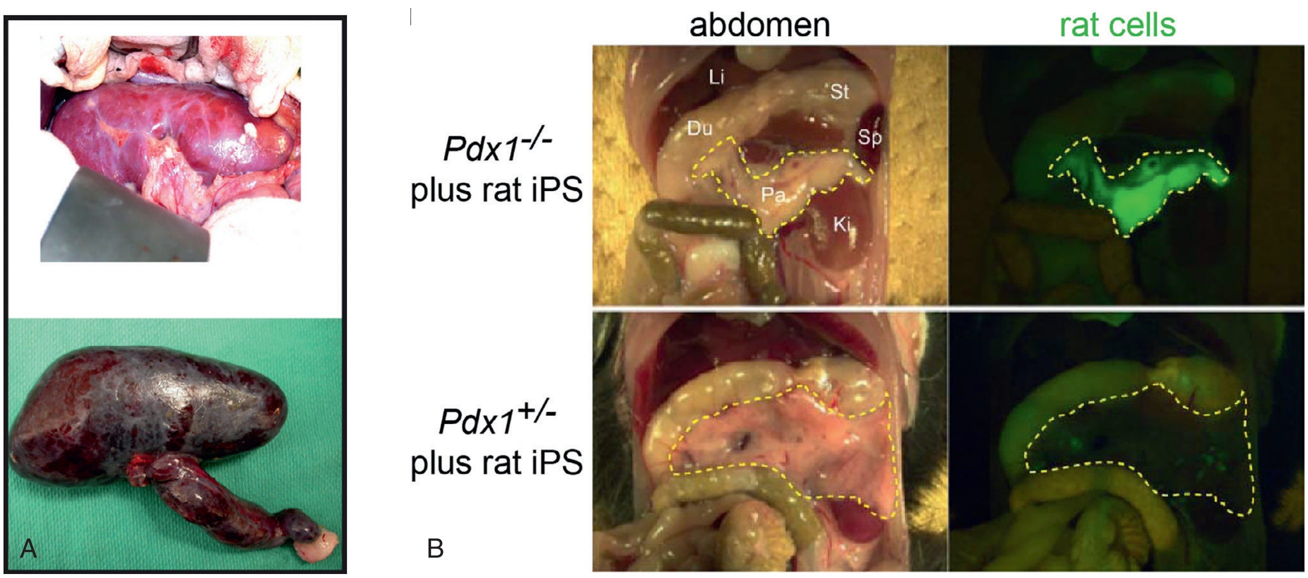

Fig. 85.5 Xenotransplantation challenges and solutions. (A) Pig kidney immediately after transplant into a baboon recipient *(top)*, and 10 minutes later after hyperacute rejection *(bottom)*. (From Cooper DK, Ezzelarab MB, Hara H, et al. The pathobiology of pig-to-primate xenotransplantation: a historical review. *Xenotransplantation* 2016;23:83–105.) (B) Rat pancreas grown in a mouse. Abdominal organs are indicated in shorthand notation. The homozygous mutant is permissive for engraftment of rat cells *(green)* to form a pancreas. (From Kobayashi T, Yamaguchi T, Hamanaka S, et al. Generation of rat pancreas in mouse by interspecific blastocyst injection of pluripotent stem cells. *Cell* 2010;142:787–799.)

Several significant technical hurdles need to be overcome, however, before IBC can be utilized successfully to farm human organs.[151] Although IBC organs are highly enriched for donor cells, the organs are still chimeric, containing a mixture of host and donor cells. One reason for this is that organs arise from many different cell populations; for instance, kidneys arise from UB, MM, nerves, vasculature, and interstitial stem cells. Genetic techniques typically establish a niche in only one of these populations, but it would be necessary to target all of them in order to create a "pure" human organ inside an animal that would not provoke an extreme immune rejection response. Another problem is that cells from one species may be poorly compatible with cells from another species during development, resulting in defects during organogenesis.

In addition to these technical concerns, human cells could potentially contribute to brains or gonads in chimeras, raising issues of human rights and dignity. Although chimerism is very low (<1%) in organs that are not targeted for complementation,[149,150] it is nevertheless important to perform these experiments in the most ethically responsible way possible. For this reason, research involving the creation of human-animal chimeric embryos is subject to a funding moratorium by the United States National Institutes of Health. Thus both ethical and technical challenges must be overcome before IBC can be tested in humans.

GENE THERAPY

Genetics contribute significantly to an individual's risk of kidney disease. Approximately 15% of kidney disease is caused directly by single-gene mutations.[152] Polycystic kidney disease (PKD), which is commonly caused by autosomal dominant mutations in either the *PKD1* or *PKD2* genes, accounts for the primary diagnosis of renal failure in ~10% of all patients.[153–155] In addition to Mendelian disorders such as PKD, a substantial portion of kidney disease genetics is complex, involving sequence variants that contribute in a combinatorial fashion to a patient's risk, such as *APOL1*.[156,157] Gene tools are furthermore emerging as powerful modulators of the immune system, with relevance to systemic kidney disorders and transplantation. A deeper understanding of disease genetics is therefore of increasing relevance for nephrology.

PREIMPLANTATION GENETIC DIAGNOSIS

Risk of hereditary kidney disease may be predictable based on family history. For instance, autosomal dominant PKD (ADPKD) is inherited as a single mutant allele from one affected parent, and therefore has a 50% chance of affecting any child of that parent. Alternatively, disease risk may be detected by prenatal screening for genetic mutations that predispose toward a variety of congenital diseases affecting the kidneys, such as Alport syndrome, autosomal recessive PKD (ARPKD), maple syrup urine disease, Fabry disease, and ciliopathy disorders.

If risk for a genetic disease is detected, prospective parents may elect to undergo preimplantation genetic diagnosis (PGD). PGD is an in vitro fertilization (IVF) technique used by parents to avoid passing on a known disease gene to their offspring. In PGD, a genetic diagnosis is made on each of several IVF embryos, based on a biopsy sample consisting of a single blastomere cell.[158] Embryos that do not carry the genetic mutation are selected for uterine implantation, while embryos with the mutation are not implanted. In this way it is possible to interrupt the transmission of disease genes from generation to generation.

Although PGD can be used to prevent any kidney disease in which genetic inheritance can be predicted, it is most commonly used for diseases that have a severe effect early in life, such as ARPKD.[159] The putative mutation that causes the disease must be known ahead of the PGD analysis, which is performed on a small sample of DNA. Sequencing of the mutation locus may be costly or require specialized techniques.[160] As PGD is a prophylactic approach, when it is successful it completely prevents the disease. However, as genotyping does not always correctly identify the mutation, the success of PGD may only become clear after the time when symptoms typically begin to present.

Clinical Relevance

Many patients who are considering starting a family may not know about PGD. It is therefore appropriate to present PGD as an option to patients that are at risk for genetic forms of kidney disease, for instance, if both of the parents are carriers of a recessive risk allele. Some patients who might otherwise choose not to have children under these circumstances might elect to have children through PGD. Others may not feel comfortable with PGD, either for religious or practical reasons, and may opt to forego it even in cases where it would be strongly recommended. Ultimately, the choice to undergo PGD is a personal one that should be made by prospective parents in an informed way, in careful consultation with their physicians.

GENE TRANSFER

In postnatal patients with genetic disease, PGD is no longer an option. Most genetic diseases result from insufficient expression levels of a functional gene. For such disorders, a straightforward strategy is to introduce a healthy copy of the gene into cells. This approach has been the cornerstone of gene therapy efforts for the past 30 years. During this time, major advances have been made with regard to the safety and efficacy of these products. Today, gene therapies are in clinical trials for a wide variety of ailments, and the United States Food and Drug Administration has approved a few of these for clinical use.

Delivery is a major challenge for all gene therapies. As small pieces of DNA are not efficiently absorbed in most tissues, successful gene therapies typically require a vector to deliver genes into cells, a process known as gene transfer. The two vectors most commonly used for gene transfer are retroviruses (including lentiviruses), which integrate into the genome and can accommodate up to 8 kb of DNA, and AAV, which are more limited in packaging size and mostly propagate as nonintegrating episomes, which have a shorter half-life than retroviral vectors. The safety and efficacy profiles of AAV and retroviral vectors are constantly improving, with advantages and disadvantages to each, depending on the context. For the kidneys, intravenous administration of lentiviral or AAV vectors is technically challenging and inefficient, at least in rodents, whereas parenchymal and retrograde (ureteral) injections may be effective if carefully performed.[161–163] AAV can be used successfully to elicit transient gene expression and is the generally preferred vector for renal gene therapy.

In certain cases, gene therapy vectors can be administered directly into the body. For instance, an AAV-based therapy for hemophilia B, encoding the factor IX gene, was administered intravenously into ten human patients, resulting in its sustained expression and the cessation of prophylactic infusions of recombinant factor IX in most of the trial subjects.[164] To date, however, most gene therapies rely upon a combination of gene and cell therapy, in which cells from the patient are first treated ex vivo and subsequently delivered back into the body. Although more laborious than direct administration, this approach overcomes certain technical hurdles in gene delivery, and presents greater opportunities for quality control checks to mitigate safety risks. Gene transfer is typically performed in adult stem cells, which can be harvested from the body and are capable of extensive self-renewal. In a recent example, large epidermal grafts were constructed from a keratinocyte clone transduced with a healthy copy of a laminin isoform, resulting in substantial remission of severe blistering disease in one patient (Fig. 85.6A).[165]

For certain renal conditions, gene therapy in an extrarenal stem cell may constitute a viable strategy to treat or prevent disease. For instance, in a mouse model of cystinosis, lentivirus gene transfer of the missing CTNS gene into HSC decreased the formation of cystine crystals in the kidneys, and moderately improved kidney function.[166] More generally, in the setting of kidney transplantation, transfer of immunomodulatory genes into HSC may improve immune tolerance of the graft. In a pig model, kidney allograft recipients first received bone marrow transplants from autologous HSC modified to express donor-specific major histocompatibility complex class II molecules, resulting in increased tolerance or acceptance of the graft in the absence of sustained immunosuppression.[167] Dendritic cells have likewise been shown to be modifiable to express immunoregulatory cytokines or receptors, which can enhance the survival of renal grafts.[168]

In contrast to skin and blood, the kidneys do not contain adult stem cells that can be modified ex vivo and subsequently used to regenerate the entire organ. Kidneys are, however, isolated ex vivo in preparation for transplant, which presents an opportunity to perform gene therapy on the entire organ. Thus gene therapy is being explored as a means of protecting donor kidneys from ischemia-reperfusion injury and immune rejection. In rat models, animal kidneys can be efficiently transduced ex vivo with AAV encoding immunomodulatory receptors, which can result in less severe outcomes after transplantation.[169–171] These studies are building towards improved graft survival rates.[9,10]

For Mendelian disorders that are intrinsic to the kidney and do not necessarily involve transplantation, such as ADPKD or Alport syndrome, gene therapy strategies remain to be tested. In addition to restoring the deficient gene, it is important to also express it at the appropriate levels, as overexpression can have deleterious effects. For instance, overexpressing *Pkd1* and *Pkd2* genes in transgenic animals can paradoxically cause PKD.[172,173] Even when employed responsibly, gene transfer carries inherent risks of side effects. For instance, retroviral gene transfer in HSC initially appeared be both safe and efficacious in treating X-linked severe combined immunodeficiency, but leukemias subsequently developed in some patients due to clonal proliferation of T-cells, in which the gene had integrated into an undesired location in the genome.[174] Cases such as these have sparked interest in more controllable, and therefore safer, gene therapy methods.

GENE EDITING

Gene editing differs from gene transfer in that it involves the deliberate alteration of a specific genetic sequence within a genome. Since the 1980s, a variety of different gene editing techniques have been developed, including site-directed homologous recombination, zinc-finger nucleases, transcription activator-like effector nucleases, piggyBac transposons, and adeno-associated viruses.[175–179] The latest and most widely

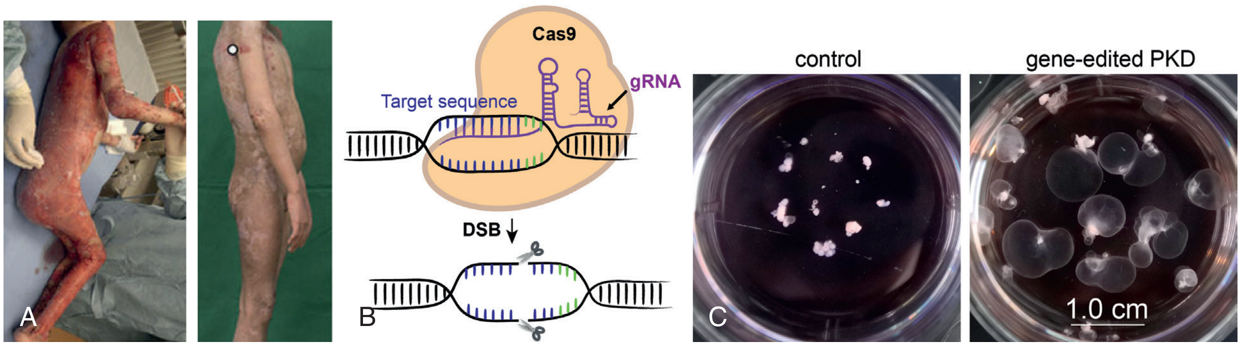

Fig. 85.6 Gene therapy and gene editing. (A) Patient with junctional epidermolysis bullosa before *(left)* and after *(right)* receiving a skin graft treated with gene therapy. White circle indicates a rare patch of non-regenerated skin. (From Hirsch T, Rothoeft T, Teig N, et al. Regeneration of the entire human epidermis using transgenic stem cells. *Nature* 2017;551:327–332.) (B) Schematic of CRISPR-Cas complex initiating DSB at a targeted site. (From Cruz NM, Freedman BS. CRISPR Gene Editing in the Kidney. *Am. J. Kidney Dis.* 2018;71(6):874–883.) (C) Kidney organoids with gene-edited PKD mutations, compared with control organoids of identical genetic background. (From Cruz NM, Song X, Czerniecki SM, et al. Organoid cystogenesis reveals a critical role of microenvironment in human polycystic kidney disease. *Nat Mater.* 2017;16:1112–1119.)

adopted of these techniques is the clustered regularly interspaced short palindromic repeats (CRISPR) system. Gene editing with this system requires the transient expression of two components: a CRISPR-associated system (Cas) endonuclease and a guide RNA (gRNA).[180,181] The gRNA combines an invariant "scaffold" sequence that binds to Cas and a variable "spacer", approximately 20 base pairs in length, which determines sequence specificity. There are four possible nucleotides for each of the positions in the spacer, providing up to 4^{20} possible sequence combinations. This exceeds the number of nucleotides in any known genome and provides sufficient sequence specificity to target unique DNA sequences complementary to the spacer.

To initiate gene editing, Cas associates with the gRNA at matching sites and introduces double-stranded breaks (DSB) in genomic DNA. Such lesions are predominantly repaired by the cell via nonhomologous end joining (NHEJ), in which the DNA ends are severed and subsequently ligated together (Fig. 85.6B). Such events frequently introduce small insertion or deletion mutations (indels). Less commonly, DSB can be corrected based on a DNA template of similar sequence, a mechanism known as homology-directed repair (HDR). This DNA template may reside elsewhere in the genome, or can be supplied artificially, providing an opportunity to introduce specific edits into DNA.[182–184]

Although CRISPR can induce DSB in a site-specific way, its effects remain relatively inefficient and unpredictable. In human PSC, for instance, the efficiency of NHEJ using CRISPR ranges from 4%–25%, and the efficiency of HDR is approximately 1% or less.[183,185] Inherent to the use of CRISPR and other gene editing methods is the risk of inadvertent edits to nuclear DNA sequences elsewhere in the genome that are similar to the target. A variety of measures can be taken to reduce the likelihood of such "off-target" effects, including the use of computational algorithms that maximize sequence specificity,[182,183] minimizing the genome's exposure to Cas, and more complex strategies that merely "nick" DNA,[186] although none of these completely eliminates the risk. Whole-genome sequencing can also be performed to verify that only the desired modifications have been made,[187,188] although this may not be economically feasible.

Despite these limitations, CRISPR and related techniques are being used to develop a new wave of gene therapies, starting with animal models. In rodent models, CRISPR has been used to inactivate pathogenic viral (HIV) or genetic (Huntington disease) sequences, by disrupting them with NHEJ.[189,190] CRISPR-induced HDR has similarly been applied to correct genetic deficiencies in muscular dystrophy[191–193] and retinal degeneration.[194–197] These studies suggest promise for application of CRISPR directly in vivo.

In many organs, including the kidneys, clinical trials have been initiated to study the potential of CRISPR-mediated gene therapy, particularly immunotherapies, which show great promise for treating certain cancers. In most of these trials, CRISPR is being used to construct T-lymphocytes that express chimeric antigen receptors (CAR-T cells), which can recognize and attack specific types of tumor cells. As such treatments can have strong side effects, they are currently restricted to patients with cancer, including metastatic renal carcinoma. Improved safety profiles might render such approaches suitable for other types of kidney diseases with an immunomodulatory component, such as lupus nephritis.[198]

CRISPR can also be applied to human embryos at the single-cell stage. Because the zygote gives rise to every other cell type in the body, a successful edit at this stage will be transmitted to the entire organism. However, due to the unpredictability of CRISPR, zygotic gene editing may be more likely to produce off-target mutations and mosaic embryos containing both modified and unmodified cells.[199–201] The efficiency of gene correction may be increased if the DSB can be repaired from a second, healthy allele, although additional studies are required to reproduce this.[184] In any case, use of CRISPR in human zygotes raises ethical concerns, and the benefits of such a therapy must outweigh the risks. PGD is almost always likely to be safer and more practical than CRISPR in human embryos, although there may be rare cases where PGD is not an option, for instance in an autosomal recessive disease where both parents are affected.

DESIGNER THERAPEUTICS

The ability to edit the genome, in combination with stem cell technology, is providing new tools for evaluation and treatment of human kidney disease. For instance, it has traditionally been challenging to study disease phenotypes in human cells, because these cultures are highly heterogeneous and do not recapitulate the complexity of the kidney. Kidney organoids provide more physiologically relevant structures for disease modeling, but PSC from patients are limited to naturally occurring mutations and exhibit substantial variability in their ability to differentiate into the kidney lineage, for reasons that are unrelated to disease.[68] Gene editing presents a possible solution to these problems by providing a way of introducing custom mutations on an otherwise identical (isogenic) genetic background. To establish a human cellular model of PKD, CRISPR was used to introduce loss-of-function mutations into *PKD1* or *PKD2* in human ES cells. Kidney organoids derived from these cells formed cysts from kidney tubules, which were not observed in isogenic controls.[50] Removal of adherent cues greatly increased the rate of cystogenesis in these organoids, resulting in their expansion into macroscopic cysts that could be observed by eye (Fig. 85.6C).[68] These experiments indicate that the essential hallmark of PKD can be reconstituted in a human cellular system in vitro, which can be used to unveil mechanisms of disease and test candidate therapeutics.

Due to the size and complexity of the human genome, inheritance mapping and genome sequencing is insufficient to conclusively identify many rare mutations as disease-causative.[202] Gene editing provides a functional strategy to rapidly validate such genes. For instance, CRISPR-mediated knockout of *GANAB*, a candidate gene for rare cases of genetically unresolved ADPKD, caused specific trafficking defects in human renal cortical tubular epithelial cells, a molecular phenotype associated with ADPKD.[203] Similarly, CRISPR knockout of *PODXL*, a candidate gene for focal segmental glomerulosclerosis,[204] caused defects in junctional migration, microvillus formation, and cell-cell spacing between podocytes in kidney organoids, similar to those observed in knockout mice.[50,69,205] Biallelic, loss-of-function mutations in *PODXL* were subsequently described in a rare patient with congenital nephrotic syndrome, consistent with the predictions of the organoid and mouse models.[206]

The use of CRISPR is not limited to the targeted editing of individual candidate genes. Genome-wide libraries of CRISPR gRNA can be applied to collections of cells in vitro, which can then be selected and analyzed for disease-relevant phenotypes.[207] Performing these screening techniques can reveal novel candidate genes for kidney disease, which can subsequently be validated in genetically unresolved family cohorts, reversing the traditional process of disease gene discovery.[208] An example of this is *PODXL*, for which the organoid phenotype preceded the case study.[69,206] Using a screening approach, modifier genes could also be identified, whose activation or inhibition can rescue kidney disease phenotypes. Noncutting variants of CRISPR-Cas have also been developed that can silence or activate genes by binding to DNA without modifying it.[209,210] In a recent example relevant to the kidney, CRISPR was used to activate expression of *Klotho* or *IL10* in the liver, which conferred renoprotection from subsequent acute kidney injury.[211]

CRISPR can also be used to more rapidly and economically manufacture regenerative therapeutics. For instance, a single zygotic injection of CRISPR can be used to generate transgenic mice and miniature pigs with mutations in multiple distinct genes.[182,212,213] This approach has been used to knock out three surface antigens that provoke acute rejection of pig cells, as well as three class I major histocompatibility complex genes, to generate pigs more immunocompatible with humans.[214,215] CRISPR has furthermore been applied in pigs to disrupt over 60 DNA sequences required for porcine endogenous retrovirus activity, resulting in virus-deficient pigs with reduced hazard of viral transmission across species lines.[216,217] Gene editing can also be combined with IBC, described earlier, to more rapidly construct chimeric embryos with organs from other species.[150]

For the kidney, a frontier is to use gene editing to produce designer human stem cells that would be immunocompatible. For instance, disease mutations could be corrected in iPS cells and rescue of phenotypes confirmed in kidney organoids in vitro, prior to implantation of NPC into a patient.[42] As generation of iPS cells and derived NPC for each individual patient may not be practical or economical,[218] an alternative strategy is to use gene editing to engineer “universal donor” cells lacking major histocompatibility genes and expressing specific immunoregulatory molecules, resulting in improved immunotolerance in humanized mice.[219,220] Although the safety and efficacy of these cells has not yet been demonstrated, such cells might provide “off-the-shelf” solutions for regenerative medicine applications with reduced need for immunosuppression. This could represent a significant improvement in the safety and availability of cell therapies for patients with kidney disease.

CONCLUSION

The survey of new technologies described earlier suggests both promise and caution with regard to renal medicine. Gene therapy and regenerative medicine remain exciting prospects, which are closer to clinical translation than ever before. In many ways, these new technologies are combinatorial and complementary—for instance, nuclear transfer techniques from the stem cell field have been critical in generating gene-edited animals for enhanced xenotransplantation. As these fields mature, efforts towards clinical translation are providing a more realistic and nuanced understanding of these technologies, revealing technical challenges and side effects that must be overcome, as illustrated by gene therapy trials. Radical changes to the practice of nephrology, where the gold standards of kidney transplantation and dialysis are well established, are therefore unlikely in the short term. To first do no harm, it is important to carefully characterize these new approaches in the research setting, before implementing them in the clinic.

With cautious and gradual adaptation of these technologies, however, it is likely that the nephrology clinic of the future will appear dramatically different from the clinic of today. The kidney has been a testing ground for many of the exciting discoveries in both the stem cell and gene editing fields and is viewed as a challenging and important frontier. New inventions, such as human kidney organoids, which did not exist only years ago, have already changed the research landscape in significant ways. Advances are being made in parallel on multiple fronts, including dialysis, transplant, and gene therapy. These accomplishments are the integrated product of decades of research by multiple groups, building toward a common vision of what care could be, rather than what it is. In practical terms, the nephrologist of the future will need to select carefully from an expanded menu of treatment options. This will require the development of new expertise with regard to stem cells, genetics, immunology, and the molecular pathophysiology. These efforts will be rewarded with increased prevention of kidney disease at early stages, and improved treatment outcomes at later stages, to prolong and enhance the lives of millions of patients.

ACKNOWLEDGMENTS

The Freedman laboratory is supported by NIH Awards K01DK102826, R01DK117914, UG3TR002158, an Allen Institute Translational Science Grant, a Regular Award from the United States–Israel Binational Science Foundation, an Institute for Stem Cell and Regenerative Medicine Innovation Pilot Award, a gift from the Northwest Kidney Centers to the Kidney Research Institute, and start-up funds from the University of Washington. Dr. Freedman is an inventor on patent applications related to human kidney organoid differentiation.

Complete reference list available at ExpertConsult.com.

KEY REFERENCES

6. Kobayashi A, Valerius MT, Mugford JW, et al. Six2 defines and regulates a multipotent self-renewing nephron progenitor population throughout mammalian kidney development. *Cell Stem Cell*. 2008;3:169–181.
7. Barak H, Huh SH, Chen S, et al. FGF9 and FGF20 maintain the stemness of nephron progenitors in mice and man. *Dev Cell*. 2012;22:1191–1207.
16. Thomson JA, Itskovitz-Eldor J, Shapiro SS, et al. Embryonic stem cell lines derived from human blastocysts. *Science*. 1998;282:1145–1147.
17. Wilmut I, Schnieke AE, McWhir J, et al. Viable offspring derived from fetal and adult mammalian cells. *Nature*. 1997;385: 810–813.
22. Wakayama S, Ohta H, Hikichi T, et al. Production of healthy cloned mice from bodies frozen at -20 degrees C for 16 years. *Proc Natl Acad Sci USA*. 2008;105:17318–17322.

23. Tachibana M, Amato P, Sparman M, et al. Human embryonic stem cells derived by somatic cell nuclear transfer. *Cell.* 2013;153:1228–1238.
26. Takahashi K, Tanabe K, Ohnuki M, et al. Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell.* 2007;131:861–872.
32. Araki R, Uda M, Hoki Y, et al. Negligible immunogenicity of terminally differentiated cells derived from induced pluripotent or embryonic stem cells. *Nature.* 2013;494:100–104.
33. Chong JJ, Yang X, Don CW, et al. Human embryonic-stem-cell-derived cardiomyocytes regenerate non-human primate hearts. *Nature.* 2014;510:273–277.
38. Mandai M, Watanabe A, Kurimoto Y, et al. Autologous induced stem-cell-derived retinal cells for macular degeneration. *N Engl J Med.* 2017;376:1038–1046.
40. Boland MJ, Hazen JL, Nazor KL, et al. Adult mice generated from induced pluripotent stem cells. *Nature.* 2009;461:91–94.
42. Freedman BS, Lam AQ, Sundsbak JL, et al. Reduced ciliary polycystin-2 in induced pluripotent stem cells from polycystic kidney disease patients with *PKD1* mutations. *J Am Soc Nephrol.* 2013;24:1571–1586.
50. Freedman BS, Brooks CR, Lam AQ, et al. Modelling kidney disease with CRISPR-mutant kidney organoids derived from human pluripotent epiblast spheroids. *Nat Commun.* 2015;6:8715.
54. Rosines E, Sampogna RV, Johkura K, et al. Staged in vitro reconstitution and implantation of engineered rat kidney tissue. *Proc Natl Acad Sci USA.* 2007;104:20938–20943.
55. Diep CQ, Ma D, Deo RC, et al. Identification of adult nephron progenitors capable of kidney regeneration in zebrafish. *Nature.* 2011;470:95–100.
59. Urbach A, Yermalovich A, Zhang J, et al. Lin28 sustains early renal progenitors and induces Wilms tumor. *Genes Dev.* 2014;28:971–982.
62. Li Z, Araoka T, Wu J, et al. 3D culture supports long-term expansion of mouse and human nephrogenic progenitors. *Cell Stem Cell.* 2016;19:516–529.
63. Taguchi A, Kaku Y, Ohmori T, et al. Redefining the in vivo origin of metanephric nephron progenitors enables generation of complex kidney structures from pluripotent stem cells. *Cell Stem Cell.* 2014;14:53–67.
64. Morizane R, Lam AQ, Freedman BS, et al. Nephron organoids derived from human pluripotent stem cells model kidney development and injury. *Nat Biotechnol.* 2015;33:1193–1200.
65. Takasato M, Er PX, Chiu HS, et al. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature.* 2016.
66. Lam AQ, Freedman BS, Morizane R, et al. Rapid and efficient differentiation of human pluripotent stem cells into intermediate mesoderm that forms tubules expressing kidney proximal tubular markers. *J Am Soc Nephrol.* 2014;25:1211–1225.
68. Cruz NM, Song X, Czerniecki SM, et al. Organoid cystogenesis reveals a critical role of microenvironment in human polycystic kidney disease. *Nat Mater.* 2017;16:1112–1119.
69. Kim YK, Refaeli I, Brooks CR, et al. Gene-edited human kidney organoids reveal mechanisms of disease in podocyte development. *Stem Cells.* 2017.
70. Taguchi A, Nishinakamura R. Higher-order kidney organogenesis from pluripotent stem cells. *Cell Stem Cell.* 2017;21:730–746 e736.
74. Hendry CE, Vanslambrouck JM, Ineson J, et al. Direct transcriptional reprogramming of adult cells to embryonic nephron progenitors. *J Am Soc Nephrol.* 2013;24:1424–1434.
76. Humphreys BD, Valerius MT, Kobayashi A, et al. Intrinsic epithelial cells repair the kidney after injury. *Cell Stem Cell.* 2008;2:284–291.
77. Rinkevich Y, Montoro DT, Contreras-Trujillo H, et al. In vivo clonal analysis reveals lineage-restricted progenitor characteristics in mammalian kidney development, maintenance, and regeneration. *Cell Rep.* 2014;7:1270–1283.
86. Appel D, Kershaw DB, Smeets B, et al. Recruitment of podocytes from glomerular parietal epithelial cells. *J Am Soc Nephrol.* 2009;20:333–343.
87. Kaverina NV, Eng DG, Schneider RR, et al. Partial podocyte replenishment in experimental FSGS derives from nonpodocyte sources. *Am J Physiol Renal Physiol.* 2016;310:F1397–F1413.
89. Lasagni L, Angelotti ML, Ronconi E, et al. Podocyte regeneration driven by renal progenitors determines glomerular disease remission and can be pharmacologically enhanced. *Stem Cell Reports.* 2015;5:248–263.
99. Duffield JS, Park KM, Hsiao LL, et al. Restoration of tubular epithelial cells during repair of the postischemic kidney occurs independently of bone marrow-derived stem cells. *J Clin Invest.* 2005;115:1743–1755.
100. Kramann R, Schneider RK, DiRocco DP, et al. Perivascular Gli1+ progenitors are key contributors to injury-induced organ fibrosis. *Cell Stem Cell.* 2015;16:51–66.
106. Imberti B, Tomasoni S, Ciampi O, et al. Renal progenitors derived from human iPSCs engraft and restore function in a mouse model of acute kidney injury. *Sci Rep.* 2015;5:8826.
107. Ronconi E, Sagrinati C, Angelotti ML, et al. Regeneration of glomerular podocytes by human renal progenitors. *J Am Soc Nephrol.* 2009;20:322–332.
110. Harari-Steinberg O, Metsuyanim S, Omer D, et al. Identification of human nephron progenitors capable of generation of kidney structures and functional repair of chronic renal disease. *EMBO Mol Med.* 2013;5:1556–1568.
111. Toyohara T, Mae S, Sueta S, et al. Cell therapy using human induced pluripotent stem cell-derived renal progenitors ameliorates acute kidney injury in mice. *Stem Cells Transl Med.* 2015;4:980–992.
117. Caralt M, Uzarski JS, Iacob S, et al. Optimization and critical evaluation of decellularization strategies to develop renal extracellular matrix scaffolds as biological templates for organ engineering and transplantation. *Am J Transplant.* 2015;15:64–75.
119. Weber EJ, Chapron A, Chapron BD, et al. Development of a microphysiological model of human kidney proximal tubule function. *Kidney Int.* 2016;90:627–637.
126. Gura V, Rivara MB, Bieber S, et al. A wearable artificial kidney for patients with end-stage renal disease. *JCI Insight.* 2016;1.
130. Humes HD, Buffington D, Westover AJ, et al. The bioartificial kidney: current status and future promise. *Pediatr Nephrol.* 2014;29:343–351.
138. Dekel B, Burakova T, Arditti FD, et al. Human and porcine early kidney precursors as a new source for transplantation. *Nat Med.* 2003;9:53–60.
139. Yokote S, Matsunari H, Iwai S, et al. Urine excretion strategy for stem cell-generated embryonic kidneys. *Proc Natl Acad Sci USA.* 2015;112:12980–12985.
143. Chen G, Qian H, Starzl T, et al. Acute rejection is associated with antibodies to non-Gal antigens in baboons using Gal-knockout pig kidneys. *Nat Med.* 2005;11:1295–1298.
147. Higginbotham L, Mathews D, Breeden CA, et al. Pre-transplant antibody screening and anti-CD154 costimulation blockade promote long-term xenograft survival in a pig-to-primate kidney transplant model. *Xenotransplantation.* 2015;22:221–230.
149. Kobayashi T, Yamaguchi T, Hamanaka S, et al. Generation of rat pancreas in mouse by interspecific blastocyst injection of pluripotent stem cells. *Cell.* 2010;142:787–799.
165. Hirsch T, Rothoeft T, Teig N, et al. Regeneration of the entire human epidermis using transgenic stem cells. *Nature.* 2017;551:327–332.
169. Tomasoni S, Trionfini P, Azzollini N, et al. AAV9-mediated engineering of autotransplanted kidney of non-human primates. *Gene Ther.* 2017;24:308–313.
181. Jinek M, Chylinski K, Fonfara I, et al. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science.* 2012;337:816–821.
216. Niu D, Wei HJ, Lin L, et al. Inactivation of porcine endogenous retrovirus in pigs using CRISPR-Cas9. *Science.* 2017;357:1303–1307.
220. Gornalusse GG, Hirata RK, Funk SE, et al. HLA-E-expressing pluripotent stem cells escape allogeneic responses and lysis by NK cells. *Nat Biotechnol.* 2017;35:765–772.

REFERENCES

1. Keller G, Zimmer G, Mall G, et al. Nephron number in patients with primary hypertension. *N Engl J Med*. 2003;348:101–108.
2. Hakim RM, Goldszer RC, Brenner BM. Hypertension and proteinuria: long-term sequelae of uninephrectomy in humans. *Kidney Int*. 1984;25:930–936.
3. Faa G, Gerosa C, Fanni D, et al. Marked interindividual variability in renal maturation of preterm infants: lessons from autopsy. *J Matern Fetal Neonatal Med*. 2010;23(suppl 3):129–133.
4. Hartman HA, Lai HL, Patterson LT. Cessation of renal morphogenesis in mice. *Dev Biol*. 2007;310:379–387.
5. Togel F, Valerius MT, Freedman BS, et al. Repair after nephron ablation reveals limitations of neonatal neonephrogenesis. *JCI Insight*. 2017;2:e88848.
6. Kobayashi A, Valerius MT, Mugford JW, et al. Six2 defines and regulates a multipotent self-renewing nephron progenitor population throughout mammalian kidney development. *Cell Stem Cell*. 2008;3:169–181.
7. Barak H, Huh SH, Chen S, et al. FGF9 and FGF20 maintain the stemness of nephron progenitors in mice and man. *Dev Cell*. 2012; 22:1191–1207.
8. Self M, Lagutin OV, Bowling B, et al. Six2 is required for suppression of nephrogenesis and progenitor renewal in the developing kidney. *EMBO J*. 2006;25:5214–5228.
9. Saran R, Robinson B, Abbott KC, et al. US Renal Data System 2016 Annual Data Report: epidemiology of kidney disease in the United States. *Am J Kidney Dis*. 2017;69:A7–A8.
10. Nankivell BJ, Alexander SI. Rejection of the kidney allograft. *N Engl J Med*. 2010;363:1451–1462.
11. Hart A, Smith JM, Skeans MA, et al. OPTN/SRTR 2016 Annual Data Report: kidney. *Am J Transplant*. 2018;18(suppl 1):18–113.
12. Hall YN, Himmelfarb J. The CKD classification system in the precision medicine era. *Clin J Am Soc Nephrol*. 2017;12:346–348.
13. Brodie JC, Humes HD. Stem cell approaches for the treatment of renal failure. *Pharmacol Rev*. 2005;57:299–313.
14. Martin GR. Isolation of a pluripotent cell line from early mouse embryos cultured in medium conditioned by teratocarcinoma stem cells. *Proc Natl Acad Sci USA*. 1981;78:7634–7638.
15. Evans MJ, Kaufman MH. Establishment in culture of pluripotential cells from mouse embryos. *Nature*. 1981;292:154–156.
16. Thomson JA, Itskovitz-Eldor J, Shapiro SS, et al. Embryonic stem cell lines derived from human blastocysts. *Science*. 1998;282:1145–1147.
17. Wilmut I, Schnieke AE, McWhir J, et al. Viable offspring derived from fetal and adult mammalian cells. *Nature*. 1997;385:810–813.
18. Gurdon JB, Elsdale TR, Fischberg M. Sexually mature individuals of Xenopus laevis from the transplantation of single somatic nuclei. *Nature*. 1958;182:64–65.
19. Liu Z, Cai Y, Wang Y, et al. Cloning of macaque monkeys by somatic cell nuclear transfer. *Cell*. 2018;172:881–887 e887.
20. Lee BC, Kim MK, Jang G, et al. Dogs cloned from adult somatic cells. *Nature*. 2005;436:641.
21. Lanza RP, Cibelli JB, Diaz F, et al. Cloning of an endangered species (Bos gaurus) using interspecies nuclear transfer. *Cloning*. 2000;2:79–90.
22. Wakayama S, Ohta H, Hikichi T, et al. Production of healthy cloned mice from bodies frozen at -20 degrees C for 16 years. *Proc Natl Acad Sci USA*. 2008;105:17318–17322.
23. Tachibana M, Amato P, Sparman M, et al. Human embryonic stem cells derived by somatic cell nuclear transfer. *Cell*. 2013;153:1228–1238.
24. Chung YG, Eum JH, Lee JE, et al. Human somatic cell nuclear transfer using adult cells. *Cell Stem Cell*. 2014;14:777–780.
25. Takahashi K, Yamanaka S. Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell*. 2006;126:663–676.
26. Takahashi K, Tanabe K, Ohnuki M, et al. Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell*. 2007;131:861–872.
27. Park IH, Zhao R, West JA, et al. Reprogramming of human somatic cells to pluripotency with defined factors. *Nature*. 2008;451:141–146.
28. Yu J, Vodyanik MA, Smuga-Otto K, et al. Induced pluripotent stem cell lines derived from human somatic cells. *Science*. 2007;318:1917–1920.
29. Hanna J, Saha K, Pando B, et al. Direct cell reprogramming is a stochastic process amenable to acceleration. *Nature*. 2009;462:595–601.
30. Bock C, Kiskinis E, Verstappen G, et al. Reference Maps of human ES and iPS cell variation enable high-throughput characterization of pluripotent cell lines. *Cell*. 2011;144:439–452.
31. Guha P, Morgan JW, Mostoslavsky G, et al. Lack of immune response to differentiated cells derived from syngeneic induced pluripotent stem cells. *Cell Stem Cell*. 2013;12:407–412.
32. Araki R, Uda M, Hoki Y, et al. Negligible immunogenicity of terminally differentiated cells derived from induced pluripotent or embryonic stem cells. *Nature*. 2013;494:100–104.
33. Chong JJ, Yang X, Don CW, et al. Human embryonic-stem-cell-derived cardiomyocytes regenerate non-human primate hearts. *Nature*. 2014;510:273–277.
34. Takebe T, Sekine K, Enomura M, et al. Vascularized and functional human liver from an iPSC-derived organ bud transplant. *Nature*. 2013;499:481–484.
35. Kroon E, Martinson LA, Kadoya K, et al. Pancreatic endoderm derived from human embryonic stem cells generates glucose-responsive insulin-secreting cells in vivo. *Nat Biotechnol*. 2008;26: 443–452.
36. Rao M. iPSC crowdsourcing: a model for obtaining large panels of stem cell lines for screening. *Cell Stem Cell*. 2013;13:389–391.
37. Grskovic M, Javaherian A, Strulovici B, et al. Induced pluripotent stem cells–opportunities for disease modelling and drug discovery. *Nat Rev Drug Discov*. 2011;10:915–929.
38. Mandai M, Watanabe A, Kurimoto Y, et al. Autologous induced stem-cell-derived retinal cells for macular degeneration. *N Engl J Med*. 2017;376:1038–1046.
39. Kamao H, Mandai M, Okamoto S, et al. Characterization of human induced pluripotent stem cell-derived retinal pigment epithelium cell sheets aiming for clinical application. *Stem Cell Reports*. 2014;2:205–218.
40. Boland MJ, Hazen JL, Nazor KL, et al. Adult mice generated from induced pluripotent stem cells. *Nature*. 2009;461:91–94.
41. Zhao XY, Li W, Lv Z, et al. iPS cells produce viable mice through tetraploid complementation. *Nature*. 2009;461:86–90.
42. Freedman BS, Lam AQ, Sundsbak JL, et al. Reduced ciliary polycystin-2 in induced pluripotent stem cells from polycystic kidney disease patients with PKD1 mutations. *J Am Soc Nephrol*. 2013;24:1571–1586.
43. Thatava T, Armstrong AS, De Lamo JG, et al. Successful disease-specific induced pluripotent stem cell generation from patients with kidney transplantation. *Stem Cell Res Ther*. 2011;2:48.
44. Xia Y, Nivet E, Sancho-Martinez I, et al. Directed differentiation of human pluripotent cells to ureteric bud kidney progenitor-like cells. *Nat Cell Biol*. 2013;15:1507–1515.
45. Chen Y, Luo R, Xu Y, et al. Generation of systemic lupus erythematosus-specific induced pluripotent stem cells from urine. *Rheumatol Int*. 2013;33:2127–2134.
46. Tropel P, Tournois J, Come J, et al. High-efficiency derivation of human embryonic stem cell lines following pre-implantation genetic diagnosis. *In Vitro Cell Dev Biol Anim*. 2010;46:376–385.
47. Teo AK, Windmueller R, Johansson BB, et al. Derivation of human induced pluripotent stem cells from patients with maturity onset diabetes of the young. *J Biol Chem*. 2013;288:5353–5356.
48. Shang L, Hua H, Foo K, et al. Beta-cell dysfunction due to increased ER stress in a stem cell model of Wolfram syndrome. *Diabetes*. 2014;63:923–933.
49. Lu S, Kanekura K, Hara T, et al. A calcium-dependent protease as a potential therapeutic target for Wolfram syndrome. *Proc Natl Acad Sci USA*. 2014;111:E5292–E5301.
50. Freedman BS, Brooks CR, Lam AQ, et al. Modelling kidney disease with CRISPR-mutant kidney organoids derived from human pluripotent epiblast spheroids. *Nat Commun*. 2015;6:8715.
51. Chen WB, Huang JR, Yu XQ, et al. Identification of microRNAs and their target genes in Alport syndrome using deep sequencing of iPSCs samples. *J Zhejiang Univ Sci B*. 2015;16:235–250.
52. Freedman BS. Modeling kidney disease with iPS cells. *Biomark Insights*. 2015;10:153–169.
53. Mugford JW, Sipila P, McMahon JA, et al. Osr1 expression demarcates a multi-potent population of intermediate mesoderm that undergoes progressive restriction to an Osr1-dependent nephron progenitor compartment within the mammalian kidney. *Dev Biol*. 2008;324:88–98.
54. Rosines E, Sampogna RV, Johkura K, et al. Staged in vitro reconstitution and implantation of engineered rat kidney tissue. *Proc Natl Acad Sci USA*. 2007;104:20938–20943.

55. Diep CQ, Ma D, Deo RC, et al. Identification of adult nephron progenitors capable of kidney regeneration in zebrafish. *Nature*. 2011;470:95–100.
56. Hohenstein P, Pritchard-Jones K, Charlton J. The yin and yang of kidney development and Wilms' tumors. *Genes Dev*. 2015;29: 467–482.
57. Beckwith JB, Kiviat NB, Bonadio JF. Nephrogenic rests, nephroblastomatosis, and the pathogenesis of Wilms' tumor. *Pediatr Pathol*. 1990;10:1–36.
58. Dekel B, Metsuyanim S, Schmidt-Ott KM, et al. Multiple imprinted and stemness genes provide a link between normal and tumor progenitor cells of the developing human kidney. *Cancer Res*. 2006;66:6040–6049.
59. Urbach A, Yermalovich A, Zhang J, et al. Lin28 sustains early renal progenitors and induces Wilms tumor. *Genes Dev*. 2014;28:971–982.
60. Brown AC, Muthukrishnan SD, Oxburgh L. A synthetic niche for nephron progenitor cells. *Dev Cell*. 2015;34:229–241.
61. Tanigawa S, Taguchi A, Sharma N, et al. Selective in vitro propagation of nephron progenitors derived from embryos and pluripotent stem cells. *Cell Rep*. 2016.
62. Li Z, Araoka T, Wu J, et al. 3D culture supports long-term expansion of mouse and human nephrogenic progenitors. *Cell Stem Cell*. 2016;19:516–529.
63. Taguchi A, Kaku Y, Ohmori T, et al. Redefining the in vivo origin of metanephric nephron progenitors enables generation of complex kidney structures from pluripotent stem cells. *Cell Stem Cell*. 2014;14:53–67.
64. Morizane R, Lam AQ, Freedman BS, et al. Nephron organoids derived from human pluripotent stem cells model kidney development and injury. *Nat Biotechnol*. 2015;33:1193–1200.
65. Takasato M, Er PX, Chiu HS, et al. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature*. 2016.
66. Lam AQ, Freedman BS, Morizane R, et al. Rapid and efficient differentiation of human pluripotent stem cells into intermediate mesoderm that forms tubules expressing kidney proximal tubular markers. *J Am Soc Nephrol*. 2014;25:1211–1225.
67. Mae S, Shono A, Shiota F, et al. Monitoring and robust induction of nephrogenic intermediate mesoderm from human pluripotent stem cells. *Nat Commun*. 2013;4:1367.
68. Cruz NM, Song X, Czerniecki SM, et al. Organoid cystogenesis reveals a critical role of microenvironment in human polycystic kidney disease. *Nat Mater*. 2017;16:1112–1119.
69. Kim YK, Refaeli I, Brooks CR, et al. Gene-edited human kidney organoids reveal mechanisms of disease in podocyte development. *Stem Cells*. 2017.
70. Taguchi A, Nishinakamura R. Higher-order kidney organogenesis from pluripotent stem cells. *Cell Stem Cell*. 2017;21:730–746 e736.
71. Sharmin S, Taguchi A, Kaku Y, et al. Human induced pluripotent stem cell-derived podocytes mature into vascularized glomeruli upon experimental transplantation. *J Am Soc Nephrol*. 2015.
72. Ieda M, Fu JD, Delgado-Olguin P, et al. Direct reprogramming of fibroblasts into functional cardiomyocytes by defined factors. *Cell*. 2010;142:375–386.
73. Vierbuchen T, Ostermeier A, Pang ZP, et al. Direct conversion of fibroblasts to functional neurons by defined factors. *Nature*. 2010;463:1035–1041.
74. Hendry CE, Vanslambrouck JM, Ineson J, et al. Direct transcriptional reprogramming of adult cells to embryonic nephron progenitors. *J Am Soc Nephrol*. 2013;24:1424–1434.
75. Kaminski MM, Tosic J, Kresbach C, et al. Direct reprogramming of fibroblasts into renal tubular epithelial cells by defined transcription factors. *Nat Cell Biol*. 2016;18:1269–1280.
76. Humphreys BD, Valerius MT, Kobayashi A, et al. Intrinsic epithelial cells repair the kidney after injury. *Cell Stem Cell*. 2008;2:284–291.
77. Rinkevich Y, Montoro DT, Contreras-Trujillo H, et al. In vivo clonal analysis reveals lineage-restricted progenitor characteristics in mammalian kidney development, maintenance, and regeneration. *Cell Rep*. 2014;7:1270–1283.
78. Oliver JA, Sampogna RV, Jalal S, et al. A subpopulation of label-retaining cells of the kidney papilla regenerates injured kidney medullary tubules. *Stem Cell Reports*. 2016;6:757–771.
79. Kusaba T, Lalli M, Kramann R, et al. Differentiated kidney epithelial cells repair injured proximal tubule. *Proc Natl Acad Sci USA*. 2014;111:1527–1532.
80. Sagrinati C, Netti GS, Mazzinghi B, et al. Isolation and characterization of multipotent progenitor cells from the Bowman's capsule of adult human kidneys. *J Am Soc Nephrol*. 2006;17:2443–2456.
81. Dor Y, Brown J, Martinez OI, et al. Adult pancreatic beta-cells are formed by self-duplication rather than stem-cell differentiation. *Nature*. 2004;429:41–46.
82. Marshall CB, Shankland SJ. Cell cycle regulatory proteins in podocyte health and disease. *Nephron Exp Nephrol*. 2007;106:e51–e59.
83. Wiggins. The spectrum of podocytopathies: a unifying view of glomerular diseases. *Kidney Int*. 2007;71:1205–1214.
84. Puelles VG, Douglas-Denton RN, Cullen-McEwen LA, et al. Podocyte number in children and adults: associations with glomerular size and numbers of other glomerular resident cells. *J Am Soc Nephrol*. 2015;26:2277–2288.
85. Vogelmann SU, Nelson WJ, Myers BD, et al. Urinary excretion of viable podocytes in health and renal disease. *Am J Physiol Renal Physiol*. 2003;285:F40–F48.
86. Appel D, Kershaw DB, Smeets B, et al. Recruitment of podocytes from glomerular parietal epithelial cells. *J Am Soc Nephrol*. 2009;20:333–343.
87. Kaverina NV, Eng DG, Schneider RR, et al. Partial podocyte replenishment in experimental FSGS derives from nonpodocyte sources. *Am J Physiol Renal Physiol*. 2016;310:F1397–F1413.
88. Wanner N, Hartleben B, Herbach N, et al. Unraveling the role of podocyte turnover in glomerular aging and injury. *J Am Soc Nephrol*. 2014.
89. Lasagni L, Angelotti ML, Ronconi E, et al. Podocyte regeneration driven by renal progenitors determines glomerular disease remission and can be pharmacologically enhanced. *Stem Cell Reports*. 2015;5:248–263.
90. Bariety J, Mandet C, Hill GS, et al. Parietal podocytes in normal human glomeruli. *J Am Soc Nephrol*. 2006;17:2770–2780.
91. Shankland SJ, Smeets B, Pippin JW, et al. The emergence of the glomerular parietal epithelial cell. *Nat Rev Nephrol*. 2014.
92. Berger K, Schulte K, Boor P, et al. The regenerative potential of parietal epithelial cells in adult mice. *J Am Soc Nephrol*. 2014;25:693–705.
93. Schulte K, Berger K, Boor P, et al. Origin of parietal podocytes in atubular glomeruli mapped by lineage tracing. *J Am Soc Nephrol*. 2014;25:129–141.
94. Shankland SJ, Freedman BS, Pippin JW. Can podocytes be regenerated in adults? *Curr Opin Nephrol Hypertens*. 2017;26:154–164.
95. Sauter A, Machura K, Neubauer B, et al. Development of renin expression in the mouse kidney. *Kidney Int*. 2008;73:43–51.
96. Pippin JW, Sparks MA, Glenn ST, et al. Cells of renin lineage are progenitors of podocytes and parietal epithelial cells in experimental glomerular disease. *Am J Pathol*. 2013;183:542–557.
97. Lichtnekert J, Kaverina NV, Eng DG, et al. Renin-angiotensin-aldosterone system inhibition increases podocyte derivation from cells of renin lineage. *J Am Soc Nephrol*. 2016;27:3611–3627.
98. Lin F, Moran A, Igarashi P. Intrarenal cells, not bone marrow-derived cells, are the major source for regeneration in postischemic kidney. *J Clin Invest*. 2005;115:1756–1764.
99. Duffield JS, Park KM, Hsiao LL, et al. Restoration of tubular epithelial cells during repair of the postischemic kidney occurs independently of bone marrow-derived stem cells. *J Clin Invest*. 2005;115:1743–1755.
100. Kramann R, Schneider RK, DiRocco DP, et al. Perivascular Gli1+ progenitors are key contributors to injury-induced organ fibrosis. *Cell Stem Cell*. 2015;16:51–66.
101. LeBleu VS, Teng Y, O'Connell JT, et al. Identification of human epididymis protein-4 as a fibroblast-derived mediator of fibrosis. *Nat Med*. 2013;19:227–231.
102. Pang P, Abbott M, Chang SL, et al. Human vascular progenitor cells derived from renal arteries are endothelial-like and assist in the repair of injured renal capillary networks. *Kidney Int*. 2017;91:129–143.
103. Kramann R, Goettsch C, Wongboonsin J, et al. Adventitial MSC-like cells are progenitors of vascular smooth muscle cells and drive vascular calcification in chronic kidney disease. *Cell Stem Cell*. 2016;19:628–642.
104. Kramann R, Wongboonsin J, Chang-Panesso M, et al. Gli1(+) pericyte loss induces capillary rarefaction and proximal tubular injury. *J Am Soc Nephrol*. 2017;28:776–784.
105. Thomas ED, Lochte HL Jr, Lu WC, et al. Intravenous infusion of bone marrow in patients receiving radiation and chemotherapy. *N Engl J Med*. 1957;257:491–496.

106. Imberti B, Tomasoni S, Ciampi O, et al. Renal progenitors derived from human iPSCs engraft and restore function in a mouse model of acute kidney injury. *Sci Rep*. 2015;5:8826.
107. Ronconi E, Sagrinati C, Angelotti ML, et al. Regeneration of glomerular podocytes by human renal progenitors. *J Am Soc Nephrol*. 2009;20:322–332.
108. Kunter U, Rong S, Boor P, et al. Mesenchymal stem cells prevent progressive experimental renal failure but maldifferentiate into glomerular adipocytes. *J Am Soc Nephrol*. 2007;18:1754–1764.
109. Kim JS, Lee JH, Kwon O, et al. Rapid deterioration of preexisting renal insufficiency after autologous mesenchymal stem cell therapy. *Kidney Res Clin Pract*. 2017;36:200–204.
110. Harari-Steinberg O, Metsuyanim S, Omer D, et al. Identification of human nephron progenitors capable of generation of kidney structures and functional repair of chronic renal disease. *EMBO Mol Med*. 2013;5:1556–1568.
111. Toyohara T, Mae S, Sueta S, et al. Cell therapy using human induced pluripotent stem cell-derived renal progenitors ameliorates acute kidney injury in mice. *Stem Cells Transl Med*. 2015;4:980–992.
112. Hitomi H, Kasahara T, Katagiri N, et al. Human pluripotent stem cell-derived erythropoietin-producing cells ameliorate renal anemia in mice. *Sci Transl Med*. 2017;9.
113. Pagliuca FW, Millman JR, Gurtler M, et al. Generation of functional human pancreatic beta cells in vitro. *Cell*. 2014;159:428–439.
114. Kuriyan AE, Albini TA, Townsend JH, et al. Vision loss after intravitreal injection of autologous "stem cells" for AMD. *N Engl J Med*. 2017;376:1047–1053.
115. Song JJ, Guyette JP, Gilpin SE, et al. Regeneration and experimental orthotopic transplantation of a bioengineered kidney. *Nat Med*. 2013;19:646–651.
116. Ross EA, Williams MJ, Hamazaki T, et al. Embryonic stem cells proliferate and differentiate when seeded into kidney scaffolds. *J Am Soc Nephrol*. 2009;20:2338–2347.
117. Caralt M, Uzarski JS, Iacob S, et al. Optimization and critical evaluation of decellularization strategies to develop renal extracellular matrix scaffolds as biological templates for organ engineering and transplantation. *Am J Transplant*. 2015;15:64–75.
118. Remuzzi A, Figliuzzi M, Bonandrini B, et al. Experimental evaluation of kidney regeneration by organ scaffold recellularization. *Sci Rep*. 2017;7:43502.
119. Weber EJ, Chapron A, Chapron BD, et al. Development of a microphysiological model of human kidney proximal tubule function. *Kidney Int*. 2016;90:627–637.
120. Jang KJ, Mehr AP, Hamilton GA, et al. Human kidney proximal tubule-on-a-chip for drug transport and nephrotoxicity assessment. *Integr Biol (Camb)*. 2013;5:1119–1129.
121. Homan KA, Kolesky DB, Skylar-Scott MA, et al. Bioprinting of 3D convoluted renal proximal tubules on perfusable chips. *Sci Rep*. 2016;6:34845.
122. Kang HW, Lee SJ, Ko IK, et al. A 3D bioprinting system to produce human-scale tissue constructs with structural integrity. *Nat Biotechnol*. 2016;34:312–319.
123. Norotte C, Marga FS, Niklason LE, et al. Scaffold-free vascular tissue engineering using bioprinting. *Biomaterials*. 2009;30:5910–5917.
124. Miller JS, Stevens KR, Yang MT, et al. Rapid casting of patterned vascular networks for perfusable engineered three-dimensional tissues. *Nat Mater*. 2012;11:768–774.
125. Ronco C, Fecondini L. The Vicenza wearable artificial kidney for peritoneal dialysis (ViWAK PD). *Blood Purif*. 2007;25:383–388.
126. Gura V, Rivara MB, Bieber S, et al. A wearable artificial kidney for patients with end-stage renal disease. *JCI Insight*. 2016;1.
127. Humes HD, Buffington DA, MacKay SM, et al. Replacement of renal function in uremic animals with a tissue-engineered kidney. *Nat Biotechnol*. 1999;17:451–455.
128. Tumlin J, Wali R, Williams W, et al. Efficacy and safety of renal tubule cell therapy for acute renal failure. *J Am Soc Nephrol*. 2008;19:1034–1040.
129. Humes HD, Weitzel WF, Bartlett RH, et al. Initial clinical results of the bioartificial kidney containing human cells in ICU patients with acute renal failure. *Kidney Int*. 2004;66:1578–1588.
130. Humes HD, Buffington D, Westover AJ, et al. The bioartificial kidney: current status and future promise. *Pediatr Nephrol*. 2014;29:343–351.
131. Fissell WH, Roy S, Davenport A. Achieving more frequent and longer dialysis for the majority: wearable dialysis and implantable artificial kidney devices. *Kidney Int*. 2013;84:256–264.
132. Buck AKW, Groszek JJ, Colvin DC, et al. Combined in silico and in vitro approach predicts low wall shear stress regions in a hemofilter that correlate with thrombus formation in vivo. *ASAIO J*. 2018;64:211–217.
133. Feinberg BJ, Hsiao JC, Park J, et al. Slit pores preferred over cylindrical pores for high selectivity in biomolecular filtration. *J Colloid Interface Sci*. 2017;517:176–181.
134. Barak H, Boyle SC. Organ culture and immunostaining of mouse embryonic kidneys. *Cold Spring Harb Protoc*. 2011;2011:pdb prot5558.
135. Abrahamson DR, St John PL, Pillion DJ, et al. Glomerular development in intraocular and intrarenal grafts of fetal kidneys. *Lab Invest*. 1991;64:629–639.
136. Rogers SA, Lowell JA, Hammerman NA, et al. Transplantation of developing metanephroi into adult rats. *Kidney Int*. 1998;54:27–37.
137. Rogers SA, Hammerman MR. Prolongation of life in anephric rats following de novo renal organogenesis. *Organogenesis*. 2004;1:22–25.
138. Dekel B, Burakova T, Arditti FD, et al. Human and porcine early kidney precursors as a new source for transplantation. *Nat Med*. 2003;9:53–60.
139. Yokote S, Matsunari H, Iwai S, et al. Urine excretion strategy for stem cell-generated embryonic kidneys. *Proc Natl Acad Sci USA*. 2015;112:12980–12985.
140. Lanza RP, Chung HY, Yoo JJ, et al. Generation of histocompatible tissues using nuclear transplantation. *Nat Biotechnol*. 2002;20:689–696.
141. Wijkstrom M, Iwase H, Paris W, et al. Renal xenotransplantation: experimental progress and clinical prospects. *Kidney Int*. 2017; 91:790–796.
142. Cowan PJ, Tector AJ. The resurgence of xenotransplantation. *Am J Transplant*. 2017.
143. Chen G, Qian H, Starzl T, et al. Acute rejection is associated with antibodies to non-Gal antigens in baboons using Gal-knockout pig kidneys. *Nat Med*. 2005;11:1295–1298.
144. Cooper DK, Ezzelarab MB, Hara H, et al. The pathobiology of pig-to-primate xenotransplantation: a historical review. *Xenotransplantation*. 2016;23:83–105.
145. Lai L, Kolber-Simonds D, Park KW, et al. Production of alpha-1,3-galactosyltransferase knockout pigs by nuclear transfer cloning. *Science*. 2002;295:1089–1092.
146. Yamada K, Yazawa K, Shimizu A, et al. Marked prolongation of porcine renal xenograft survival in baboons through the use of alpha1,3-galactosyltransferase gene-knockout donors and the cotransplantation of vascularized thymic tissue. *Nat Med*. 2005;11:32–34.
147. Higginbotham L, Mathews D, Breeden CA, et al. Pre-transplant antibody screening and anti-CD154 costimulation blockade promote long-term xenograft survival in a pig-to-primate kidney transplant model. *Xenotransplantation*. 2015;22:221–230.
148. Iwase H, Liu H, Wijkstrom M, et al. Pig kidney graft survival in a baboon for 136 days: longest life-supporting organ graft survival to date. *Xenotransplantation*. 2015;22:302–309.
149. Kobayashi T, Yamaguchi T, Hamanaka S, et al. Generation of rat pancreas in mouse by interspecific blastocyst injection of pluripotent stem cells. *Cell*. 2010;142:787–799.
150. Wu J, Platero-Luengo A, Sakurai M, et al. Interspecies chimerism with mammalian pluripotent stem cells. *Cell*. 2017;168:473–486 e415.
151. Freedman BS. Hopes and difficulties for blastocyst complementation. *Nephron*. 2017.
152. McKnight AJ, Currie D, Maxwell AP. Unravelling the genetic basis of renal diseases; from single gene to multifactorial disorders. *J Pathol*. 2010;220:198–216.
153. Gilg J, Methven S, Casula A, et al. UK Renal Registry 19th Annual Report: Chapter 1 UK RRT adult incidence in 2015: national and centre-specific analyses. *Nephron*. 2017;137(suppl 1):11–44.
154. The polycystic kidney disease 1 gene encodes a 14 kb transcript and lies within a duplicated region on chromosome 16. The European Polycystic Kidney Disease Consortium. *Cell*. 1994;78:725.
155. Mochizuki T, Wu G, Hayashi T, et al. PKD2, a gene for polycystic kidney disease that encodes an integral membrane protein. *Science*. 1996;272:1339–1342.
156. Tzur S, Rosset S, Shemer R, et al. Missense mutations in the APOL1 gene are highly associated with end stage kidney disease risk previously attributed to the MYH9 gene. *Hum Genet*. 2010;128:345–350.
157. Genovese G, Friedman DJ, Ross MD, et al. Association of trypanolytic ApoL1 variants with kidney disease in African Americans. *Science*. 2010;329:841–845.

158. Vermeesch JR, Voet T, Devriendt K. Prenatal and pre-implantation genetic diagnosis. *Nat Rev Genet*. 2016;17:643–656.
159. De Rycke M, Georgiou I, Sermon K, et al. PGD for autosomal dominant polycystic kidney disease type 1. *Mol Hum Reprod*. 2005; 11:65–71.
160. Zeevi DA, Renbaum P, Ron-El R, et al. Preimplantation genetic diagnosis in genomic regions with duplications and pseudogenes: long-range PCR in the single-cell assay. *Hum Mutat*. 2013;34:792–799.
161. Gusella GL, Fedorova E, Hanss B, et al. Lentiviral gene transduction of kidney. *Hum Gene Ther*. 2002;13:407–414.
162. Espana-Agusti J, Tuveson DA, Adams DJ, et al. A minimally invasive, lentiviral based method for the rapid and sustained genetic manipulation of renal tubules. *Sci Rep*. 2015;5:11061.
163. Konkalmatt PR, Asico LD, Zhang Y, et al. Renal rescue of dopamine D2 receptor function reverses renal injury and high blood pressure. *JCI Insight*. 2016;1.
164. Miesbach W, Meijer K, Coppens M, et al. Gene therapy with adeno-associated virus vector 5-human factor IX in adults with hemophilia B. *Blood*. 2017.
165. Hirsch T, Rothoeft T, Teig N, et al. Regeneration of the entire human epidermis using transgenic stem cells. *Nature*. 2017;551: 327–332.
166. Harrison F, Yeagy BA, Rocca CJ, et al. Hematopoietic stem cell gene therapy for the multisystemic lysosomal storage disorder cystinosis. *Mol Ther*. 2013;21:433–444.
167. Sonntag KC, Emery DW, Yasumoto A, et al. Tolerance to solid organ transplants through transfer of MHC class II genes. *J Clin Invest*. 2001;107:65–71.
168. Gorczynski RM, Bransom J, Cattral M, et al. Synergy in induction of increased renal allograft survival after portal vein infusion of dendritic cells transduced to express TGFbeta and IL-10, along with administration of CHO cells expressing the regulatory molecule OX-2. *Clin Immunol*. 2000;95:182–189.
169. Tomasoni S, Trionfini P, Azzollini N, et al. AAV9-mediated engineering of autotransplanted kidney of non-human primates. *Gene Ther*. 2017;24:308–313.
170. Yang J, Reutzel-Selke A, Steier C, et al. Targeting of macrophage activity by adenovirus-mediated intragraft overexpression of TNFRp55-Ig, IL-12p40, and vIL-10 ameliorates adenovirus-mediated chronic graft injury, whereas stimulation of macrophages by overexpression of IFN-gamma accelerates chronic graft injury in a rat renal allograft model. *J Am Soc Nephrol*. 2003;14:214–225.
171. Benigni A, Tomasoni S, Turka LA, et al. Adeno-associated virus-mediated CTLA4Ig gene transfer protects MHC-mismatched renal allografts from chronic rejection. *J Am Soc Nephrol*. 2006;17:1665–1672.
172. Pritchard L, Sloane-Stanley JA, Sharpe JA, et al. A human PKD1 transgene generates functional polycystin-1 in mice and is associated with a cystic phenotype. *Hum Mol Genet*. 2000;9:2617–2627.
173. Park EY, Sung YH, Yang MH, et al. Cyst formation in kidney via B-Raf signaling in the PKD2 transgenic mice. *J Biol Chem*. 2009;284:7214–7222.
174. Hacein-Bey-Abina S, Von Kalle C, Schmidt M, et al. LMO2-associated clonal T cell proliferation in two patients after gene therapy for SCID-X1. *Science*. 2003;302:415–419.
175. Smithies O, Gregg RG, Boggs SS, et al. Insertion of DNA sequences into the human chromosomal beta-globin locus by homologous recombination. *Nature*. 1985;317:230–234.
176. Thomas KR, Folger KR, Capecchi MR. High frequency targeting of genes to specific sites in the mammalian genome. *Cell*. 1986;44:419–428.
177. Liu Q, Segal DJ, Ghiara JB, et al. Design of polydactyl zinc-finger proteins for unique addressing within complex genomes. *Proc Natl Acad Sci USA*. 1997;94:5525–5530.
178. Cermak T, Doyle EL, Christian M, et al. Efficient design and assembly of custom TALEN and other TAL effector-based constructs for DNA targeting. *Nucleic Acids Res*. 2011;39:e82.
179. Russell DW, Hirata RK. Human gene targeting by viral vectors. *Nat Genet*. 1998;18:325–330.
180. Gasiunas G, Barrangou R, Horvath P, et al. Cas9-crRNA ribonucleoprotein complex mediates specific DNA cleavage for adaptive immunity in bacteria. *Proc Natl Acad Sci USA*. 2012;109:E2579–E2586.
181. Jinek M, Chylinski K, Fonfara I, et al. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*. 2012;337:816–821.
182. Cong L, Ran FA, Cox D, et al. Multiplex genome engineering using CRISPR/Cas systems. *Science*. 2013;339:819–823.
183. Mali P, Yang L, Esvelt KM, et al. RNA-guided human genome engineering via Cas9. *Science*. 2013;339:823–826.
184. Ma H, Marti-Gutierrez N, Park SW, et al. Correction of a pathogenic gene mutation in human embryos. *Nature*. 2017;548:413–419.
185. Ding Q, Regan SN, Xia Y, et al. Enhanced efficiency of human pluripotent stem cell genome editing through replacing TALENs with CRISPRs. *Cell Stem Cell*. 2013;12:393–394.
186. Ran FA, Hsu PD, Lin CY, et al. Double nicking by RNA-guided CRISPR Cas9 for enhanced genome editing specificity. *Cell*. 2013; 154:1380–1389.
187. Veres A, Gosis BS, Ding Q, et al. Low incidence of off-target mutations in individual CRISPR-Cas9 and TALEN targeted human stem cell clones detected by whole-genome sequencing. *Cell Stem Cell*. 2014;15:27–30.
188. Smith C, Gore A, Yan W, et al. Whole-genome sequencing analysis reveals high specificity of CRISPR/Cas9 and TALEN-based genome editing in human iPSCs. *Cell Stem Cell*. 2014;15:12–13.
189. Yin C, Zhang T, Qu X, et al. In vivo excision of HIV-1 provirus by saCas9 and multiplex single-guide RNAs in animal models. *Mol Ther*. 2017;25:1168–1186.
190. Yang S, Chang R, Yang H, et al. CRISPR/Cas9-mediated gene editing ameliorates neurotoxicity in mouse model of Huntington's disease. *J Clin Invest*. 2017;127:2719–2724.
191. Long C, Amoasii L, Mireault AA, et al. Postnatal genome editing partially restores dystrophin expression in a mouse model of muscular dystrophy. *Science*. 2016;351:400–403.
192. Nelson CE, Hakim CH, Ousterout DG, et al. In vivo genome editing improves muscle function in a mouse model of Duchenne muscular dystrophy. *Science*. 2016;351:403–407.
193. Tabebordbar M, Zhu K, Cheng JKW, et al. In vivo gene editing in dystrophic mouse muscle and muscle stem cells. *Science*. 2016; 351:407–411.
194. Bakondi B, Lv W, Lu B, et al. In vivo CRISPR/Cas9 gene editing corrects retinal dystrophy in the S334ter-3 rat model of autosomal dominant retinitis pigmentosa. *Mol Ther*. 2016;24:556–563.
195. Latella MC, Di Salvo MT, Cocchiarella F, et al. In vivo editing of the human mutant rhodopsin gene by electroporation of plasmid-based CRISPR/Cas9 in the mouse retina. *Mol Ther Nucleic Acids*. 2016;5:e389.
196. Suzuki K, Tsunekawa Y, Hernandez-Benitez R, et al. In vivo genome editing via CRISPR/Cas9 mediated homology-independent targeted integration. *Nature*. 2016;540:144–149.
197. Yu W, Mookherjee S, Chaitankar V, et al. Nrl knockdown by AAV-delivered CRISPR/Cas9 prevents retinal degeneration in mice. *Nat Commun*. 2017;8:14716.
198. Cruz NM, Freedman BS. CRISPR gene editing in the kidney. *Am J Kidney Dis*. 2018.
199. Kang X, He W, Huang Y, et al. Introducing precise genetic modifications into human 3PN embryos by CRISPR/Cas-mediated genome editing. *J Assist Reprod Genet*. 2016;33:581–588.
200. Liang P, Xu Y, Zhang X, et al. CRISPR/Cas9-mediated gene editing in human tripronuclear zygotes. *Protein Cell*. 2015;6:363–372.
201. Tang L, Zeng Y, Du H, et al. CRISPR/Cas9-mediated gene editing in human zygotes using Cas9 protein. *Mol Genet Genomics*. 2017;292:525–533.
202. McClellan J, King MC. Genetic heterogeneity in human disease. *Cell*. 2010;141:210–217.
203. Porath B, Gainullin VG, Cornec-Le Gall E, et al. Genkyst Study Group HPoPKDG, Consortium for Radiologic Imaging Studies of Polycystic Kidney D, Harris PC: mutations in GANAB, encoding the glucosidase IIalpha subunit, cause Autosomal-Dominant polycystic kidney and liver disease. *Am J Hum Genet*. 2016;98:1193–1207.
204. Barua M, Shieh E, Schlondorff J, et al. Exome sequencing and in vitro studies identified podocalyxin as a candidate gene for focal and segmental glomerulosclerosis. *Kidney Int*. 2014;85:124–133.
205. Doyonnas R, Kershaw DB, Duhme C, et al. Anuria, omphalocele, and perinatal lethality in mice lacking the CD34-related protein podocalyxin. *J Exp Med*. 2001;194:13–27.
206. Kang HG, Lee M, Lee KB, et al. Loss of podocalyxin causes a novel syndromic type of congenital nephrotic syndrome. *Exp Mol Med*. 2017;49:e414.
207. Shalem O, Sanjana NE, Hartenian E, et al. Genome-scale CRISPR-Cas9 knockout screening in human cells. *Science*. 2014;343:84–87.
208. Breslow DK, Hoogendoorn S, Kopp AR, et al. A CRISPR-based screen for Hedgehog signaling provides insights into ciliary function and ciliopathies. *Nat Genet*. 2018.

209. Konermann S, Brigham MD, Trevino AE, et al. Genome-scale transcriptional activation by an engineered CRISPR-Cas9 complex. *Nature*. 2015;517:583–588.
210. Qi LS, Larson MH, Gilbert LA, et al. Repurposing CRISPR as an RNA-guided platform for sequence-specific control of gene expression. *Cell*. 2013;152:1173–1183.
211. Liao HK, Hatanaka F, Araoka T, et al. In vivo target gene activation via CRISPR/Cas9-mediated trans-epigenetic modulation. *Cell*. 2017;171:1495–1507 e1415.
212. Wang H, Yang H, Shivalila CS, et al. One-step generation of mice carrying mutations in multiple genes by CRISPR/Cas-mediated genome engineering. *Cell*. 2013;153:910–918.
213. Wang X, Cao C, Huang J, et al. One-step generation of triple gene-targeted pigs using CRISPR/Cas9 system. *Sci Rep*. 2016;6:20620.
214. Estrada JL, Martens G, Li P, et al. Evaluation of human and non-human primate antibody binding to pig cells lacking GGTA1/CMAH/beta4GalNT2 genes. *Xenotransplantation*. 2015;22:194–202.
215. Reyes LM, Estrada JL, Wang ZY, et al. Creating class I MHC-null pigs using guide RNA and the Cas9 endonuclease. *J Immunol*. 2014;193:5751–5757.
216. Niu D, Wei HJ, Lin L, et al. Inactivation of porcine endogenous retrovirus in pigs using CRISPR-Cas9. *Science*. 2017;357:1303–1307.
217. Yang L, Guell M, Niu D, et al. Genome-wide inactivation of porcine endogenous retroviruses (PERVs). *Science*. 2015;350:1101–1104.
218. Turner M, Leslie S, Martin NG, et al. Toward the development of a global induced pluripotent stem cell library. *Cell Stem Cell*. 2013;13:382–384.
219. Rong Z, Wang M, Hu Z, et al. An effective approach to prevent immune rejection of human ESC-derived allografts. *Cell Stem Cell*. 2014;14:121–130.
220. Gornalusse GG, Hirata RK, Funk SE, et al. HLA-E-expressing pluripotent stem cells escape allogeneic responses and lysis by NK cells. *Nat Biotechnol*. 2017;35:765–772.

BOARD REVIEW QUESTIONS

1. A patient with stable chronic kidney disease (CKD) resulting from focal segmental glomerulosclerosis arrives at the clinic with sudden and severe acute kidney injury (AKI). Renal biopsy findings at dialysis show severe interstitial fibrosis and inflammatory cell infiltration, with the unusual appearance of fat globules within the glomerular tufts. These findings could be explained by which of the following events in the weeks prior to admission?
   a. Mutations in *PODXL* resulting in transdifferentiation of podocytes into adipose tissue
   b. Increased cholesterol consumption in training for a competitive eating competition
   c. Intravenous administration of autologous mesenchymal stem cells (MSC) from a rogue stem cell clinic
   d. Adeno-associated viral gene therapy targeting congenital nephrotic syndrome of the Finnish type

   Answer: c

   Rationale: Rogue stem cell clinics frequently offer MSC treatments for a variety of diseases, including kidney disorders. Autologous MSC can be readily purified from adipose tissue. Following injection, these stem cells can survive and differentiate into ectopic connective tissues, such as fat. This can cause severe and sudden worsening of kidney symptoms in patients with CKD.

2. True or false: mammalian nephron progenitor cells (NPC) are never observed to persist naturally after birth.
   a. True
   b. False

   Answer: b

   Rationale: Although NPC do not normally persist after birth, there are rare examples of cases where they do persist, such as Wilms tumor.

3. A 3-year-old patient with Wilms tumor is treated with an experimental therapy consisting of CAR-T cells, which induces remission of the cancer. Four years later, the patient is diagnosed with a proliferative, clonal T-cell lymphoma. The CAR-T cells used in the original therapy were most likely prepared using which of the following methods?
   a. Retroviral gene transfer
   b. Adenoviral gene transfer
   c. CRISPR-Cas nonhomologous end joining
   d. CRISPR-Cas homology-directed repair

   Answer: a

   Rationale: Retroviruses integrate randomly into the genome and can disrupt genetic elements that control the expression of oncogenes. Failure to properly characterize gene therapies can result in the introduction of pro-oncogenic cell lines into the body. In contrast, adenoviruses are predominantly nonintegrating, whereas CRISPR is targeted to specific loci in the genome.

4. A patient with a familial history of autosomal dominant polycystic kidney disease elects to perform preimplantation genetic diagnosis (PGD) prior to having children. Healthy fraternal twins are conceived by IVF and successfully delivered. Thirty years later, a routine ultrasound reveals numerous bilateral cysts in one of the twins, while the other twin is found to have no cysts. Which of the following is the most likely reason for recurrence of polycystic kidney disease (PKD)?
   a. Human error in handling the samples taken from the biopsied embryos
   b. A spontaneous somatic mutation in nephron progenitor cells occurring *in utero*
   c. Inaccurate diagnosis of the genetic mutation
   d. Failure of CRISPR to properly initiate homology-directed repair from the homologous chromosome

   Answer: c

   Rationale: The chances of inheriting ADPKD are typically 50%. Preimplantation genetic diagnosis reduces this risk by selecting embryos that do not carry likely mutations. However, due to the complexities of human genetics, the precise mutation that causes PKD may not be fully predictable based on sequencing data. Erroneous diagnosis of the disease-causative mutation renders the PGD ineffective, therefore the children inherit the disease in the classic Mendelian ratios for an autosomal dominant disorder. CRISPR is not used in PGD, and somatic mutations in developing kidneys or human error in handling expensive PGD samples would be expected to be a much rarer occurrence.